



CDS505: Data Visualisations & Visual Analytics

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Group Project

Group 01

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List of Abbreviation

EduTech	Educational Technology
TAM	Technology Acceptance Model

1.0 Introduction and background

Data Story Telling Website Link: <https://zw7.github.io/cds505-studentperformance/>

1.1 Background

The relationship between student effort and academic achievement has long been assumed to follow a straightforward pattern, study more, achieve more. In reality, student academic success is influenced by multiple factors, including behavioural, psychological, and other factors beyond ability. In recent years, student well-being has been increasingly recognised as a significant factor in learning outcomes, rather than only on academic results. Students' stress levels, motivation, class participation, and so on are actually closely associated with persistence, retention, and achievement, regardless of face-to-face or online learning environment. In a deeper layer of discussion, it is important to study the linkage of students' engagement in coursework and or group study to stress level. Such complex multifactorial influences are difficult for educators, parents, or decision makers to use a simple metric to gauge, in order to make meaningful decisions. Therefore, visualizing and contextualizing the causality of different contexts with academic performances is important and addressed in this project.

1.2 Dataset Description and Preparation

Dataset: <https://www.kaggle.com/datasets/adilshamim8/student-performance-and-learning-style>

The dataset was obtained from Kaggle, containing 14,000 anonymized student records with 16 attributes covering demographics, learning behaviour, and academic performance. It integrates academic performance indicators with different behavioural and psychological attributes, together with demographic data, which reflects student engagement and well-being. It encompasses a range of variables such as stress level, motivation level, participation in class discussions, assignment completion, exam scores, and many more, in a total of 16 attributes. Hence, this is why this dataset was selected in this project, as these features enable multidimensional exploration of different contexts that correlate to academic success.

For dataset preparation, we performed data filtering to remove outliers and anomalous records that did not reflect realistic student behaviour. For instance, records with extreme values such as unrealistic study hours or inconsistent grade-score combinations were excluded to ensure the visualisations generate meaningful insights applicable to real-world educational contexts. After this data cleaning process, the final dataset comprises about 9000 anonymized student records, which forms the foundation for all subsequent visual analytics in this project.

1.3 Target audience

The deliverables generated from this project are intended to support data exploration, information presentation, and decision-making, as well as improve the understanding of several related parties:

- a. *Educators*: who can identify at-risk students and behavioural patterns that may benefit from personal-level intervention actions, like counselling and additional support.
- b. *Researchers*: who seek empirical insights into how well-being, environment, engagement, or other factors might correlate to academic metrics.
- c. *Educational administrators and decision makers*: who require evidence to form policies on sustaining and improving the entire educational system of the area.
- d. *Individuals like students or parents*: who can use some visualized context, for self-reflection on habits that are linked to performance outcomes of students.

1.4 Related work

Research on student performance consistently shows that academic outcomes are influenced by multiple factors, rather than by any single variable. This comprehensive view necessitates a deep exploration of the identified predictors to fully understand and address variations in student success.

A core area of research focuses on the internal characteristics and behaviors of students. Wang and Fan (2025) identified that learning motivation, engagement, time management and mental health are key predictive factors with assignment completion quality significantly affecting the student performance. Researchers had found that intrinsic motivation is strongly associated with higher achievement as a student (Howard et al., 2021), but some research also found that stress and anxiety affect student ability to complete assignments and participate in class, leading to reduced performance (Pascoe et al., 2019). These findings lead our focus on exploring the relationships among different factors that will affect the student performance such as completion, student profile, technology access, motivation and stress level.

Moreover, with the contemporary educational landscape, that had increasing reliance on online and blended learning, had amplified the significance of technology access as a performance determinant. The unequal distribution of technological resources and digital access is often referred to as the digital divide which creates systemic inequality in academic outcomes. Subramaniam et al. (2024) examined the digital divide among Malaysian university students, finding that technology disparities affected online learning outcomes. Akpen et al. (2024) demonstrated that variations in student engagement substantially impact performance in online environments. Hurtado-Arhuata et al. (2024) established a significant relationship between technology resource quality and academic performance, highlighting how infrastructure limitations disadvantage certain student groups. These studies validate our task abstraction focusing on technology access as a factor associated with academic equity.

Interactive dashboards had been proven effective for monitoring student performance by previous research. Amira et al. (2024) developed a Power BI dashboard for student academic performance analysis, demonstrating high user acceptance through Technology Acceptance Model (TAM) evaluation. From a theoretical perspective, Munzner (2014) provides the framework for our encoding decisions and task abstractions, while Schneiderman's (1996) visual information-seeking mantra informs our interaction design.

Based on these related works, our project will integrate the educational domain knowledge with the visualization principles from Munzner (2014) to create an interactive data story telling website. Unlike prediction focused approaches, we prioritize interpretability through coordinated multiple views, which enables stakeholders to discover actionable relationships among different factors that could affect student performance.

2. Data Abstraction

2.1 Data Items

Each data item in this dataset represents an individual student record, encompassing demographic, behavioural, psychological, and academic information. Unlike time-series data, this dataset is cross-sectional, meaning each data item reflects a snapshot of a student's academic and its correlated context within a single, simplified academic period.

2.2 Attributes

This dataset consists of a combination of categorical and quantitative attributes:

Attributes	Type	Description
StudyHours	Quantitative	Number of study hours per week
Attendance	Quantitative	Percentage of classes attended
Resources	Categorical	Availability and use of academic resources
Extracurricular	Categorical	Participation in extracurricular activities
Motivation	Ordered	Self-reported motivation level
Internet	Categorical	Access to the internet for study purposes
Gender	Categorical	Gender of the student
Age	Quantitative	Age of student
LearningStyle	Categorical	Preferred learning style
OnlineCourses	Categorical	Participation in online courses
Discussions	Categorical	Engagement in study group discussions or forums
AssignmentCompletion	Ordered	Rate of completing assignments on time, on a scale
ExamScore	Quantitative	Score obtained in the main exam
EduTech	Categorical	Usage of educational technology tools or platforms
StressLevel	Ordered	Self-reported stress level
FinalGrade	Ordered	Final course grade

2.3 Dataset Types

The dataset type is a **table**. In some sense, it is a multidimensional table that can be separated into behavioural, psychological, environmental, engagement, and so on. However, we can only obtain a single table in a spreadsheet file from Kaggle, and there is no unique identifier attached, so it is just a simple multivariate table. Where each row represents a student, and each column represents an attribute. From a visualization perspective, this structure is well-suited for aggregate visualizations such as a tree map and bubble chart, where grouping and comparison are primary analytic goals.

2.4 Derived attribute

Among the attributes, there are several derived attributes used to support the visualization, rather than applied directly, including:

Derived Attribute	Description
Correlation Coefficients	Calculated from raw attributes to show relationship strength between each variable and FinalGrade.
Average values by Grade Level	Mean StudyHours and mean AssignmentCompletion calculated per grade category such as Excellent, Good, Passed, Failed.
Student Counts per Grade	Aggregated counts derived by grouping and counting records per FinalGrade category.
Binned Exam Scores	Exam scores grouped into bins for the various visualizations to have a more detailed view on certain situation where using performance band is not significant enough..
Normalized Values of Attributes	Raw attribute values are normalised to a common scale from 0 to 1 so that different metrics with different units can be compared on the same radar axes.

3. Task Abstraction

Following Munzner's framework, task abstraction focuses on **why** a visualization exists by specifying the actions users need to perform and the targets of those actions. This makes sure that each visualization serves a clear analytical purpose. In other words, this part focuses on explaining in what way the action and target pair of this visual representation is able to help the target user carry out the task more effectively. Our analysis identified 4 tasks that users must accomplish when exploring student performance factors.

3.1 Task 1: Identify Key Predictors of Academic Success

The main action of this task is **analysis**, with the objective of **discovering** previously unknown relationships through data **consumption**. Users engage with the visual representations to examine how different variables contribute to academic performance, rather than approaching the data with a fixed hypothesis. By consuming the data through multiple coordinated views, users are encouraged to **explore** which factors exhibit the strongest associations with student outcomes. To successfully perform this analytical task, the visualizations must support the discovery of meaningful patterns, trends, and distributions that would be difficult to detect through tabular data alone.

The targets of this task operate across several levels of abstraction. At the highest level, the **attribute** target encompasses overall exam scores and final grade outcomes, providing essential context for understanding the general distribution of student performance. This global view allows users to form an initial mental model of how achievement is spread across the population. At a more detailed level, the task focuses on a set of fifteen individual variables and their relationships with performance levels, enabling users to **compare** how different academic, behavioural, and contextual factors align with outcomes. In addition, a derived **feature** target is introduced through the computation of correlation coefficients between each variable and final grade allowing users to assess the relative strength and direction of each variable's relationship with academic performance.

The purpose of this task is to address a central question in educational analysis: which factors demonstrate the strongest relationship with student academic performance? By doing so, the task challenges the widely held assumption that longer study hours alone are the primary driver of better grades. While this assumption strongly influences student behaviour and institutional expectations, it may overlook other influential factors such as engagement, motivation, or completion behaviours. Through this task abstraction, the visualization provides evidence-based insights that help stakeholders **identify** attributes with empirical support, rather than relying on intuition or convention. These insights can inform more effective decision making related to resource allocation, academic support, and targeted interventions.

To support this analytical goal, the visualization design employs four linked visualizations within a compact mini dashboard. These include a histogram of exam scores to convey overall performance distribution, a grade distribution bar chart to **summarize** categorical outcomes, a grouped bar chart to compare average completion metrics across grade categories, and a correlation heatmap to reveal predictor and student performance outcome relationships.

3.2 Task 2: Compare Student Profiles Across Performance Levels

The purpose of this task is to analyze with the goal to compare student profiles across performance level by consuming aggregated summaries. In this task, users first **discover** overall differences by scanning group summaries across the performance categories. The performance category is divided into four which is Excellent, Good, Passed, Failed; where Excellent corresponds to Grade A and Failed corresponds to Grade F. Users then **compare** the same attribute across performance groups to determine the direction and size of differences. After comparing, users **identify** the attributes that best separate higher-performing and lower-performing students, and **summarize** each performance level as a clear typical profile that can be described and communicated.

The main target is the performance groups created from the final outcome categories. The comparison is applied to a defined set of **attributes** representing student behaviors and conditions. For example, StudyHours, Attendance, AssignmentCompletion, OnlineCourses, ExamScore, plus additional attributes shown in the profile view such as Discussions, Motivation, StressLevel, Resources, EduTech, Internet, and Extracurricular. These attributes together form a profile **pattern** for each performance group. In this design, the profile pattern is represented using normalized values in the radar chart so users can read the overall pattern consistently across attributes, while the grouped bar chart presents selected attributes in their original units to support clear within-attribute group comparison.

This task consists of three linked analytical steps. First, users **compare** group summaries by checking how each attribute changes across the performance groups. For example, comparing Attendance across Excellent, Good, Passed, Failed, then comparing AssignmentCompletion across Excellent, Good, Passed, Failed as well. Second, users **identify** distinguishing characteristics by focusing on attributes with the clearest separation between the top and bottom groups, rather than treating all attributes as equally important. Third, users optionally **locate** outliers by enabling the outlier display to inspect exceptions that do not match the typical group pattern, supporting cautious interpretation rather than assuming the profile applies to every student.

This task is supported by two complementary visualisation designs aligned to the **comparison** goal. The grouped bar chart supports direct group comparison for selected key attributes. The grouped bar chart is intended for comparing groups within the same attribute. However, it is not for comparing different attributes against each other, since attributes have different units. The radar chart supports profile comparison by showing each performance group as a single normalized pattern across multiple attributes, which supports a concise profile summary for each group. The outlier toggle supports additional inspection by allowing users to focus on subsets and to check unusual cases without changing the core comparison structure.

3.3 Task 3: Explore the Impact of Learning Environment & Technology

In education analytics, technology access is often discussed as part of the digital divide where limited access to the internet, devices, or learning platforms lead to different learning outcomes. Previous studies in higher education contexts including Malaysia, indicates that digital divide conditions can affect students' online learning experiences and involvement (Subramaniam et al., 2024). The domain problem is to determine how educational technology access and online learning engagement relate to student academic performance, and whether these relationships vary among learning styles. The visualization is then designed to help in analytical reasoning.

To address this by following Munzner's framework, specific domain questions such as "Does education technology improve student exam scores?", are abstracted into generic visualization tasks that can be efficiently supported by visual encodings. This method ensures that each visualization directly helps the users achieve their goal without being too complicated.

The visualization supports users to **discover** and **compare** academic performance trends across different levels of online learning engagement. This work focuses on determining the **trend** and stability of exam scores as the number of online courses increases while at the same time considering education technology access. Users can **analyze** and discover how exam scores vary with increasing number of online course participation. Besides, users can figure out whether performance increases, decreases, or fluctuates, and the different patterns between students with and without access to educational technology. This task provides a high level overview of the dataset and creates context for subsequent comparisons. The visualization shows that students with EduTech access tend to score better in exams and have more stable performance, whereas students without EduTech access tend to have more varied exam scores.

The visualization supports users to **identify** learning style groups that are most affected by technology access. This is about figuring out which groups of learners are most affected by differences in technology access and users may also look at the average test scores for each learning style in both high-tech and low-tech conditions. By **comparing** score differences within each learning style category, users can **identify** which groups gain the most from technology access and which groups may be disadvantaged when technology access is limited. Bubble size gives users more information about how well each group is represented, which helps them consider the number of students belonging to each category. Through the visualization, users can observe that some learning styles such as Kinaesthetic and Visual show the largest performance difference between high and low technology situations. In contrast, other learning styles show smaller performance gaps which are low sensitivity to technology availability.

Moreover, this task helps users to **summarize** the overall findings from the visualization and determine whether the findings are consistent. Users can check how reliable insights are by looking at how stable the **trends** are and whether similar **patterns** occur in different views. This helps users avoid focusing on small and not important changes and better assess how reliable the results are for implementation.

3.4 Task 4: Analyze Well-being and Engagement Factors

The primary action of this task is to **analyze**, with the aim of discovering relationships between student well-being and engagement factors and academic performance outcomes. In this part, we try to **explore** unknown knowledge behind different attributes. Users **consume** the data to examine how stress levels, motivation, extracurricular participation, and class discussions relate to exam performance and overall academic achievement.

The targets of this task operate at multiple levels. At the **attribute** level, the task focuses on academic performance indicators such as exam scores and performance categories. At the factor level, the task examines well-being and engagement attributes, including stress level, motivation, and participation-related variables. These targets allow users to reason about how different conditions and behaviours correspond to variations in academic outcomes, rather than evaluating performance in isolation.

The purpose of this task is to address the domain question of how student well-being and engagement influence academic performance. It challenges simplified assumptions that academic success is driven solely by cognitive effort or study time, by highlighting the role of psychological and behavioural factors. Through this task abstraction, stakeholders such as educators, researchers, students, and parents can gain a more holistic understanding of academic performance drivers, supporting evidence-based interventions aimed at improving both student well-being and learning outcomes.

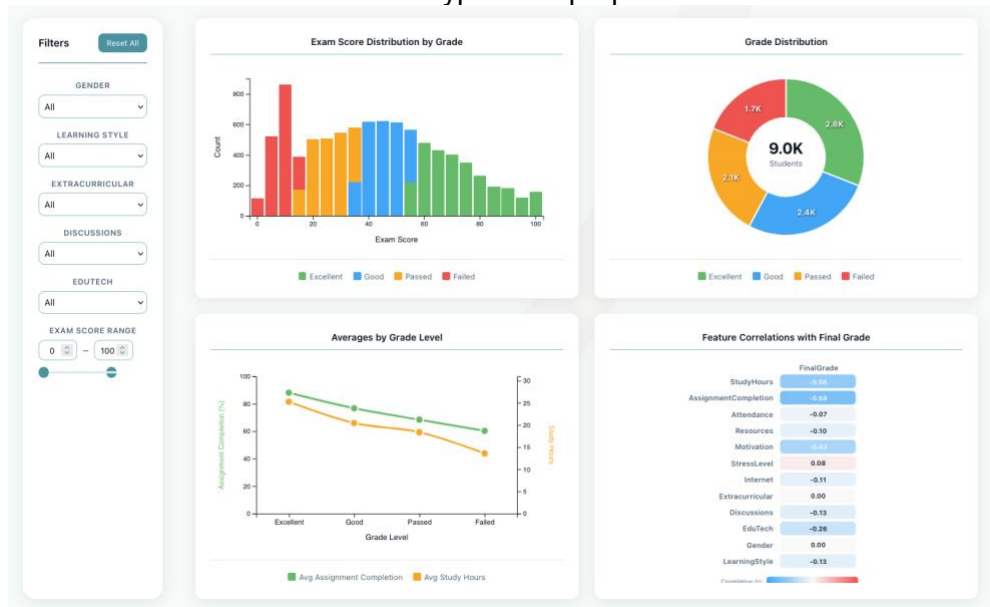
This task is effective because it supports multiple analytical intents within a single coherent objective, and hence multiple linked visualizations are designed as usual. It enables users to **discover** patterns in how stress relates to performance distribution, analyze how motivation interacts with academic outcomes under varying conditions, and **summarize** the overall effect of engagement-related behaviours on performance. By integrating supplementary discovery, relational analysis, and summarization within the same task, users are able to build progressively deeper insights, moving from pattern identification to higher-level reasoning about contributing factors and potential trade-offs affecting academic success. However, at the highest level, the primary target is still **analysed** since there is a clear target of knowledge we intend to extract.

4. Design Justification

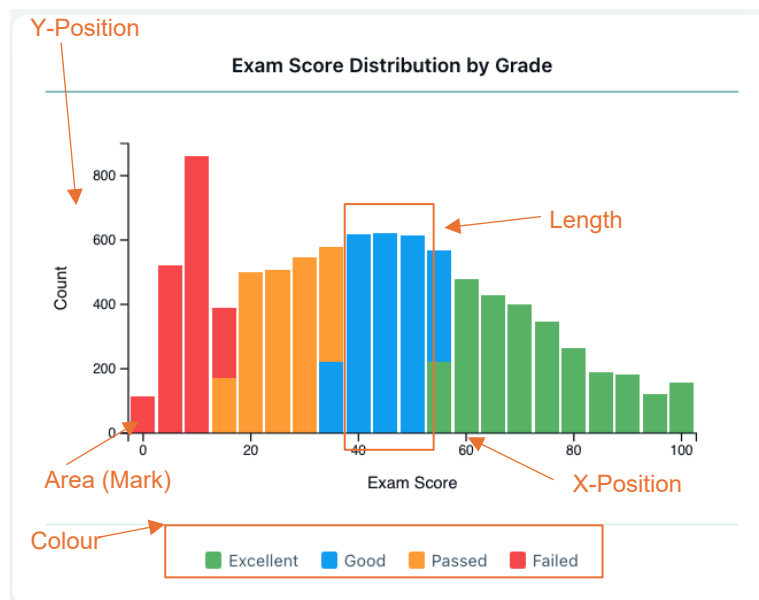
4.1 Visualizations for Task 1

4.1.1 Marks and Channel

The Task 1 mini dashboard employs four linked visualisations, each utilizing appropriate marks and channels based on their data types and purposes.



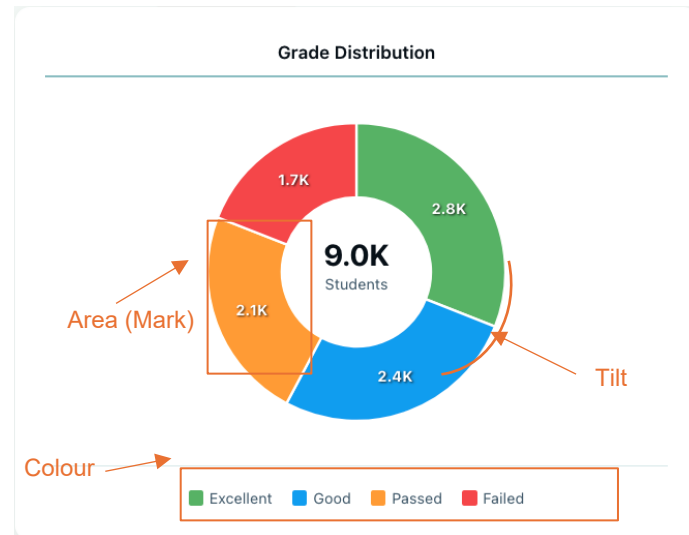
The Exam Score Distribution by Grade, histogram uses *area* marks with position on the horizontal *x-axis* encoding exam score bins and *length* on the vertical *y-axis* encoding student count. *Colour hue* distinguishes the four grade categories: green for Excellent, blue for Good, orange for Passed, and red for Failed. This stacked arrangement enables users to perceive both overall distribution shape and grade composition within each score range.



Marks: Area (Rectangular Bars)

Channels: X-Position, Y-Position, Colour Hue, Length

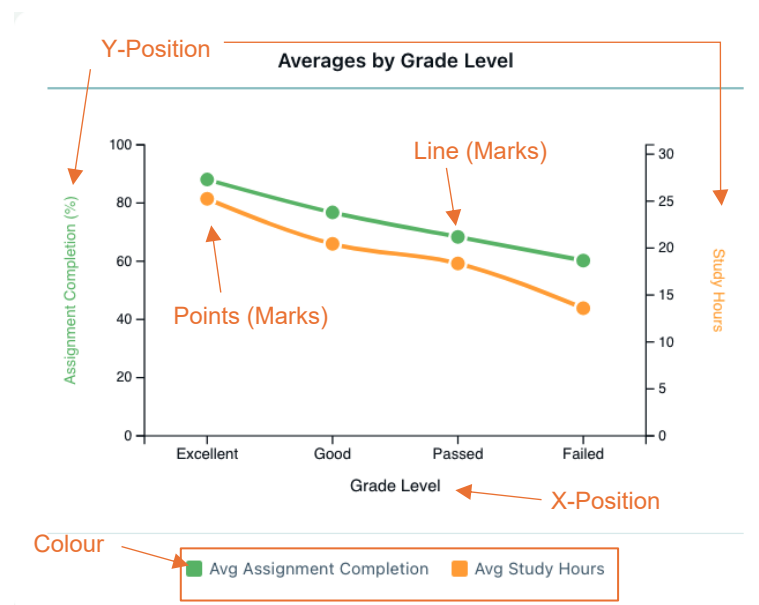
The Grade Distribution, donut chart uses *area* marks with *angles* encoding the proportion of students in each grade category. *Colour hue* maintains consistency with the histogram, using the same four-colour scheme. The central label displays the total count, providing immediate context for the proportional segments showing the number of students with Excellent, Good, Passed and Failed.



Marks: Area

Channels: Colour Hue, Tilt (Angle)

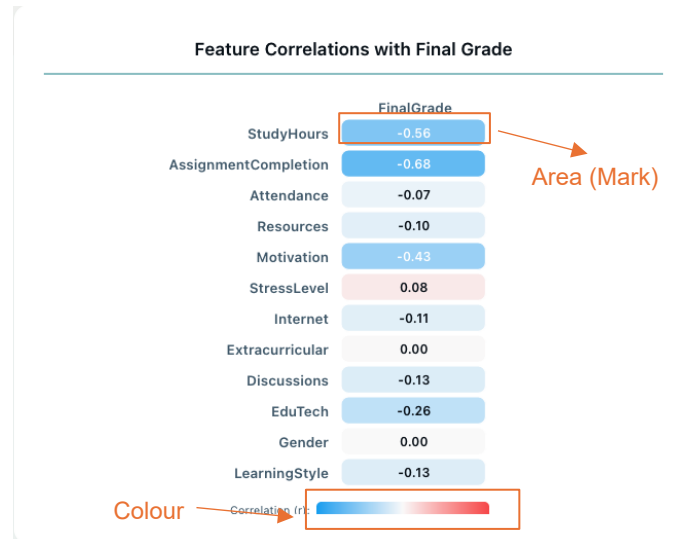
The Averages by Grade Level, line chart that uses *point* marks connected by *line* marks, with position on the horizontal x-axis encoding grade categories ordered from Excellent to Failed and position on the vertical y-axis encoding metric values. Two separate *colour hues* distinguish the metrics, green for Average Assignment Completion at the *left y-axis* and orange for Average Study Hours at the *right y-axis*, hours scale. This dual-axis design enables direct comparison of how both metrics decline across performance levels.



Marks: Line, Points

Channels: X-Position, Y-Position (Left), Y-Position (Right), Colour Hue

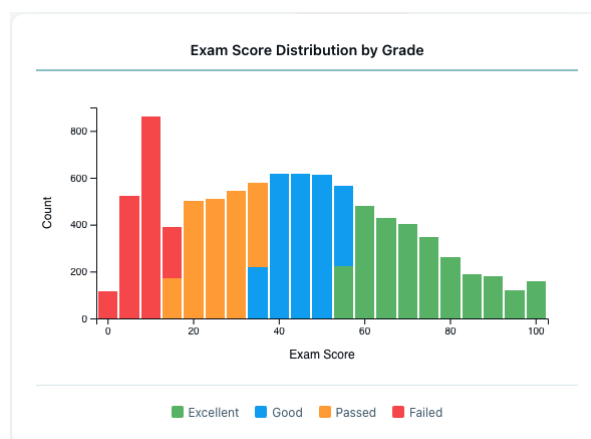
The Feature Correlations with Final Grade, heatmap uses *area* marks with position encoding variable names on the vertical axis. *Colour hue* and saturation encode correlation coefficient values using a diverging scheme, blue for negative correlations which indicates better performance and red for positive correlations, with intensity representing correlation strength. Numeric labels show the correlation coefficient values for the users.



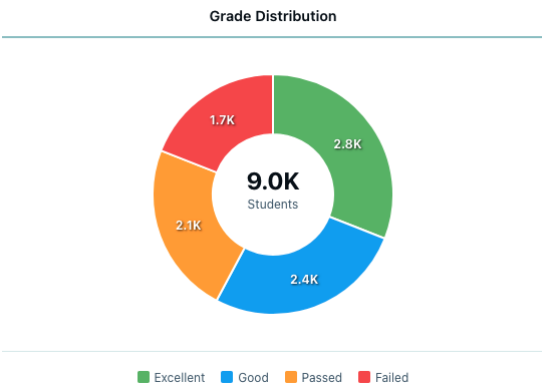
Marks: *Area (Rectangular Cells)*
Channels: *Colour Hue*

4.1.2 Visual Encoding and Analytics

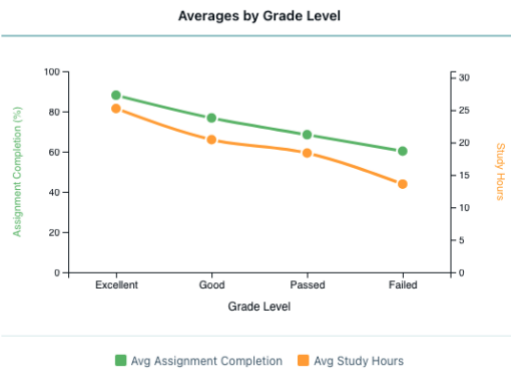
The visual encoding decisions of this task follow the Munzner's (2014) principles of expressiveness and effectiveness. The most effective channel for quantitative data, positions are utilized to encode the primary comparisons such as exam scores in histogram, grade categories in the line chart and variable ordering in the heatmap. Colour hues which are the most effective for categorical data are consistently used to identify grade categories across visualizations using the same colour schemes, reducing the cognitive load as users shift attention between different views.



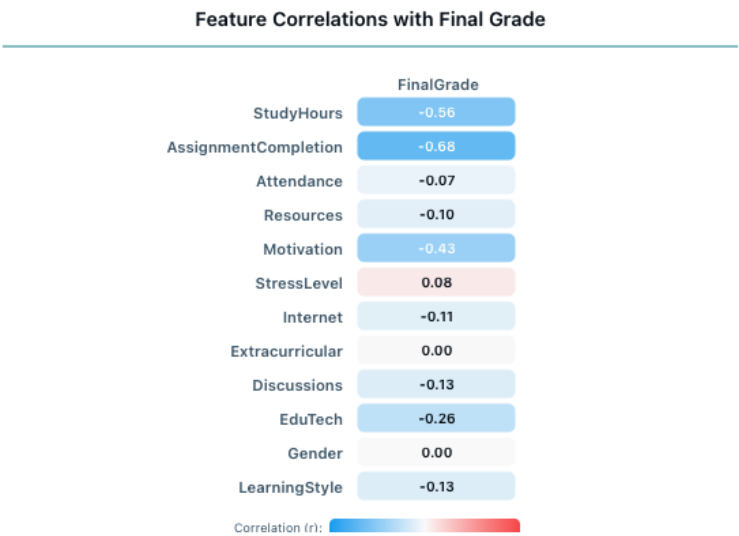
The histogram with stacked colour encoding enables two analytical operations simultaneously. The first operation is the users can identify the overall distribution shape of exam score while comparing how grade categories occupy the distinct score regions. The clear separation reveals that excellent students identified by green colour hue cluster in higher score ranges while failed students identified by red colour hue concentrate in lower range of the exam scores.



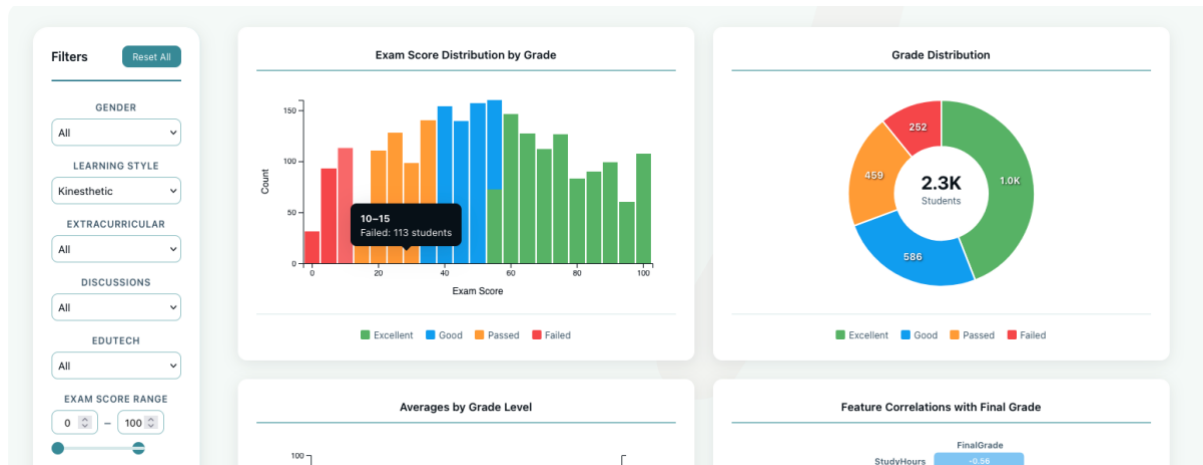
Besides that, the donut chart provides rapid summarisation of categorical outcomes through angle encoding, showing the number of counts of Excellent, Good, Passed, and Failed students. The central label of total students provides immediate context for proportion estimation.



Moreover, the dual-axis line chart reveals a critical insight through position encoding. While both metrics decline from Excellent to Failed, assignment completion shows steeper decline compared to study hours. This visual pattern supports the discovery that completion differentiates performance more strongly than time invested.



Furthermore, the correlation heatmap uses colour saturation to encode the correlation strength which immediately highlighting that assignment completion and study hours shows strongest relationships with the final grade of a student, while gender and extracurricular show minimal relationships. The diverging colour scheme enables quick identification of high-impact predictors for the users.

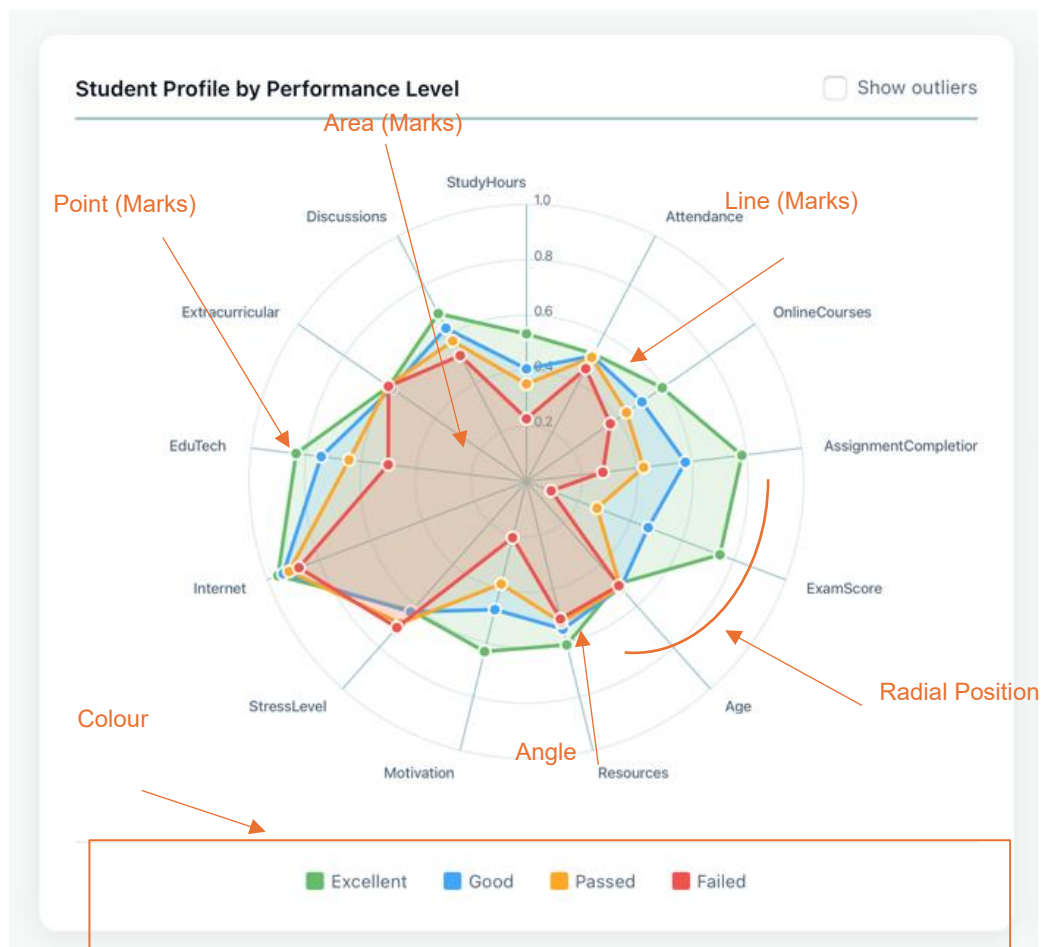


The task 1 mini dashboard also implements the four interaction types following Ben Schneiderman's influential mantra of Overview First, Zoom and Filter, Details on Demand (1996). Dropdown filters for Gender, Learning Style, Extracurricular, Discussions, and EduTech implement the select interaction for subset reduction on a large dataset which enables focus and context while reducing cognitive load. The exam score range slider allows filter interaction, querying for reduction to enable smooth exploration of continuous attributes. All visualizations for this task feature hover tooltips that allow for users to interact for the details, providing precise values on demand to reduce visual clutter while enabling deeper exploration. Finally, the slicer filter affects all four charts in the task 1 mini dashboard simultaneously, following the coordinated multiple views pattern that increases confidence in discovered patterns across different representations.

4.2 Visualizations for Task 2

4.2.1 Marks and Channel

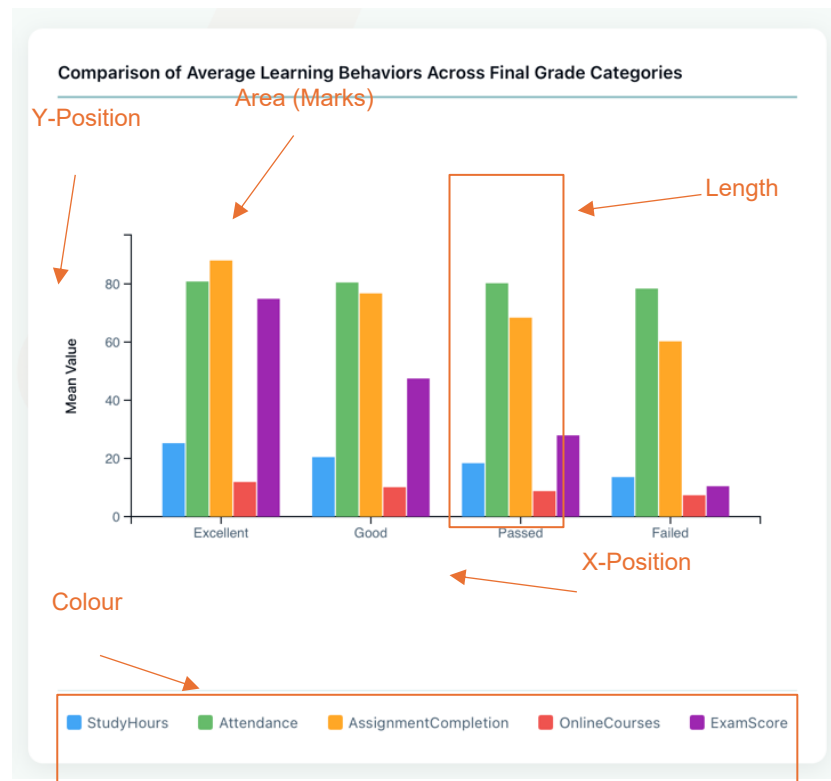
The task 2 uses complementary visualisation designs to support group comparison and profile reading which is a radar chart for multivariate profile patterns and a grouped bar chart for precise within attribute group comparison. The radar chart encodes each performance level, which is Excellent, Good, Passed, Failed as a separate profile line around a circular axis. Attributes as shown in the radar chart below are arranged by angular position, and the normalized mean value for each attribute is encoded by radial distance from the center (0 - 1 scale). Colour hue will distinguish the four performance groups, and point markers on each axis in order to help users to see the group mean at each attribute. The filled polygon area will then reinforces the overall student profile pattern for quick scanning.



Marks: *Line, Area (filled polygon), Point*

Channels: *Angle (attribute), Radial position (normalized mean value), Colour hue (performance group)*

The grouped bar chart encodes the performance group on the axis and mean values on the y-axis. Each selected attribute is represented by a separate coloured bar within each performance group, enabling side by side comparison of the attribute across groups. Colour hue encodes attributes identity, for example StudyHours, Attendance, AssignmentCompletion, OnlineCourses, ExamScore and also the bar length encodes magnitude.



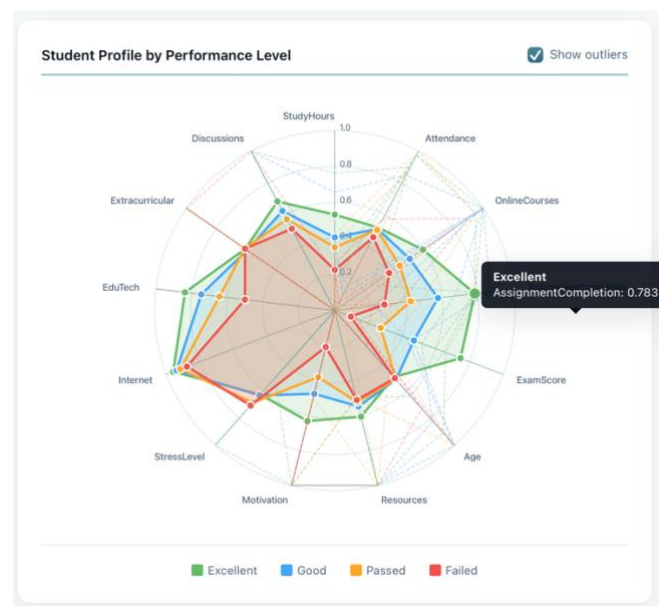
Marks: Area (rectangular bars)

Channels: X-Position (performance group), Y-Position (mean value), Bar Length (mean value), Colour hue (attribute)

4.2.2 Visual Encoding and Analytics

The visual encoding for Task 2 is designed to support the actions in Section 3.2, where users discover differences, compare performance groups on the same attribute, identify the most separating attributes, and summarize each group as a typical profile. The design in this task has two visualisation designs. The first design is the radar chart supports profile reading across many attributes at once, which makes it efficient for discovering where performance groups diverge and for summarizing an overall pattern for each group. The second visualisation design in task two is the grouped bar chart supports more precise comparison on a smaller set of key attributes, where users can compare groups on a single attribute with high accuracy and quickly identify the attributes with the clearest gaps.

In the radar chart, each performance level is represented as a single multivariate profile line drawn across all selected attributes. Attributes are placed around the circle, and the mean value for each attribute is encoded by radial distance from the center. Because the attributes have different original units for example, hours, percentages, and scores, values are normalized to a common 0–1 scale so the overall profile pattern is comparable across axes. This encoding supports rapid pattern reading which users can discover that higher performing groups extend further outward on attributes such as AssignmentCompletion and ExamScore, while lower performing groups remain closer to the center on these axes. At the same time, attributes such as Attendance show relatively small separation between groups, which helps prevent over interpreting variables that appear stable across performance levels.



The grouped bar chart encodes performance level on the x-axis and mean value on the y-axis, with a separate coloured bar for each selected attribute within each performance group. This visualisation design is intended for within attribute comparison where users compare the same attribute across Excellent, Good, Passed, and Failed using bar length and aligned position, which provides strong perceptual support for judging differences. This makes it straightforward to identify the characteristics between top and bottom groups, such as the strong gap in AssignmentCompletion and ExamScore. In the meantime, the graph also confirms that Attendance varies only slightly across categories. The design also supports careful interpretation by discouraging cross attribute comparison within the same group. Since the attributes remain in their own original units and bar height is only meaningful when comparing the same attribute across groups.



Interactivity is used to support details on demand and exception checking without cluttering the default display. Hover provides details on demand by revealing the exact group value for the selected mark for example performance level, attribute name, and mean, which supports confirmation after users discover an apparent difference and reduces the need to label all values directly on the chart. By turning on the outlier toggle and adding dashed lines in the same time to the radar chart so that users can easily see cases that sit outside the typical group profile. This makes it easier to identify students who do not follow the usual pattern for their performance level. For example, someone with unusually high online course activity or unusually low motivation compared with the group average. By combining with the hover function, all of these interactions are able to make the charts more trustworthy and interesting. The main profile lines show the overall trend, while the interactive features let users check exact values and investigate exceptions without making the view hard to read.

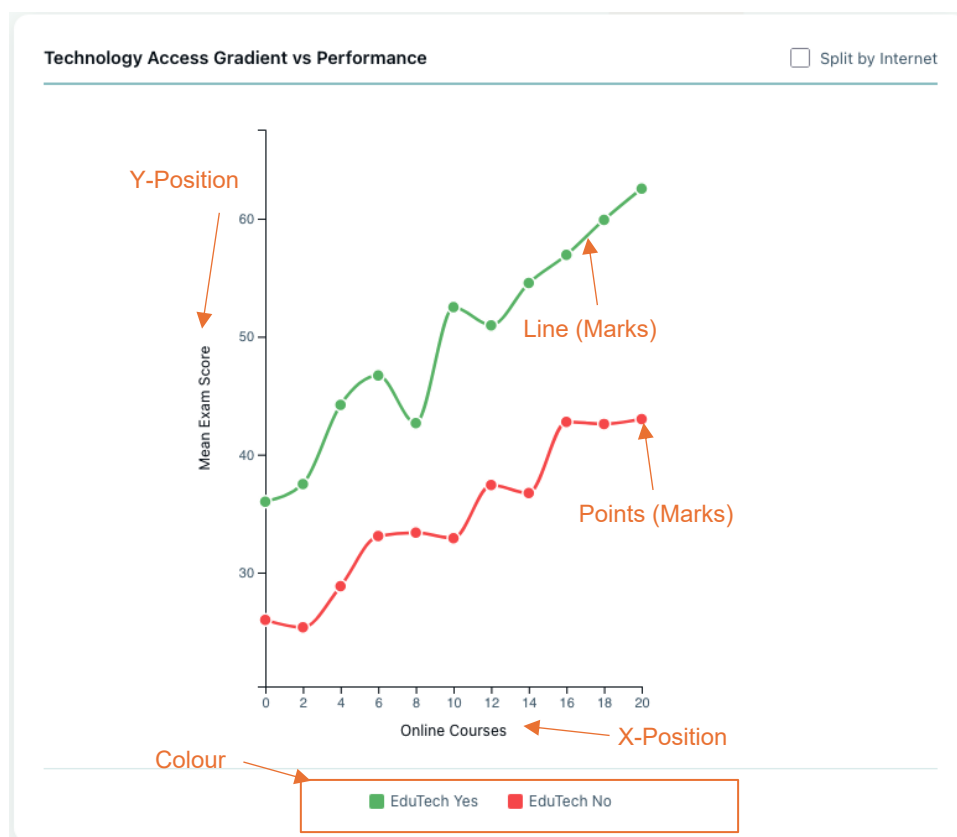
Overall, these encoding choices support what this analysis is trying to do in task two. The radar chart gives a quick, normalized snapshot of each group's typical profile across many attributes, which helps with discovery and summarizing patterns. The grouped bar chart then makes it easier to compare performance groups on specific key behaviours with more precision, so differences are clearer and easier to identify for users. Hover tooltips and the outlier overlay add a useful layer of checking, users can easily confirm exact values and look at unusual cases without cluttering the main view.

4.3 Visualizations for Task 3

4.3.1 Marks and Channel

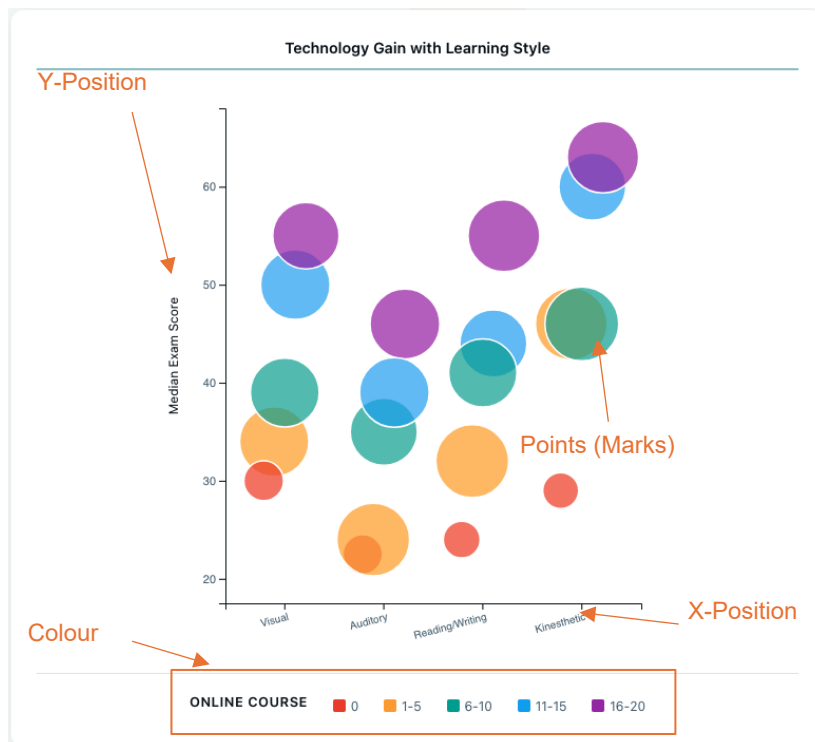
The visualization uses two primary types of visual marks, each applied differently across the two charts. In the line chart, items are represented by lines, where the horizontal axis (x-axis) represents online course participation, while the vertical axis (y-axis) shows exam scores. Colour hue is used as the main channel to differentiate between students with and without education technology access that enables users directly make comparisons between two groups. For the bubble chart, items are represented as points. Each bubble represents a learning style and is positioned along the horizontal axis (x-axis), while vertical position (y-axis) shows the average exam score. Colour hue is used to encode technology access level (high tech and low tech) which allows the users to compare performance within each learning style. Bubble size indicates group representation by providing additional context regarding the number of students in each category. The marks and channels used in Task 3 visualization as follows:

Line chart



Mark: *Line, Point*

Channels: *X-Position, Y-Position, Colour Hue*

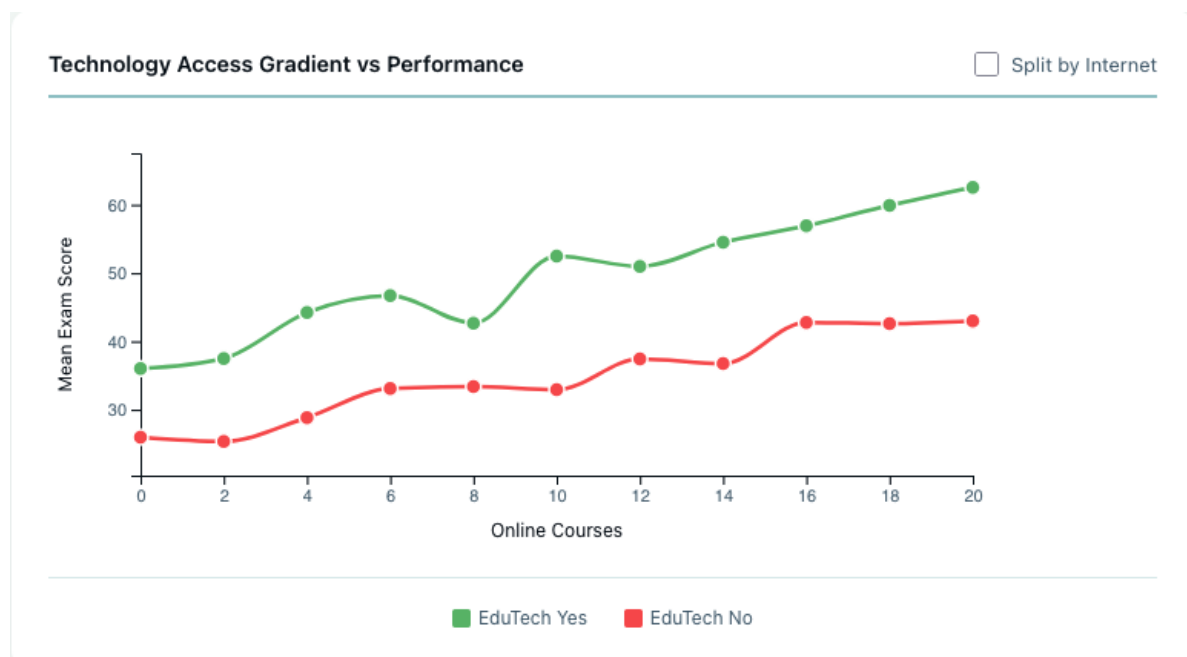
Bubble chart

Mark: *Point*

Channel: *X-Position, Y-Position, Colour Hue, Size*

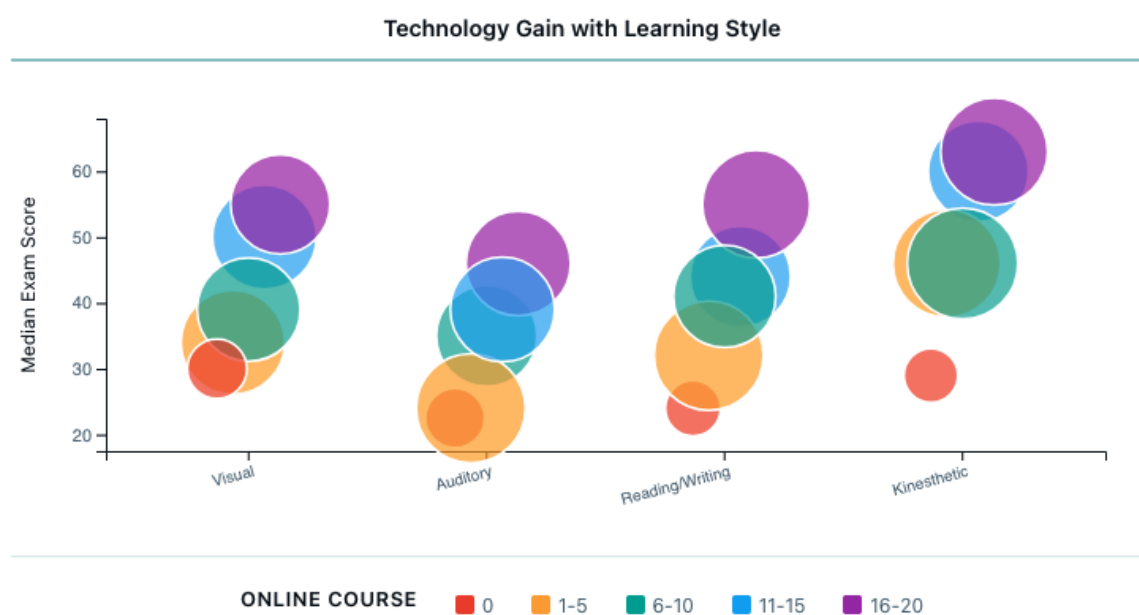
4.3.2 Visual Encoding and Analytics

The visual encodings are designed to support analytical reasoning tasks such as discovering trends, identifying affected segments, and comparing groups. These tasks are simplified by the combination of position, colour, and size across the line chart and bubble chart.



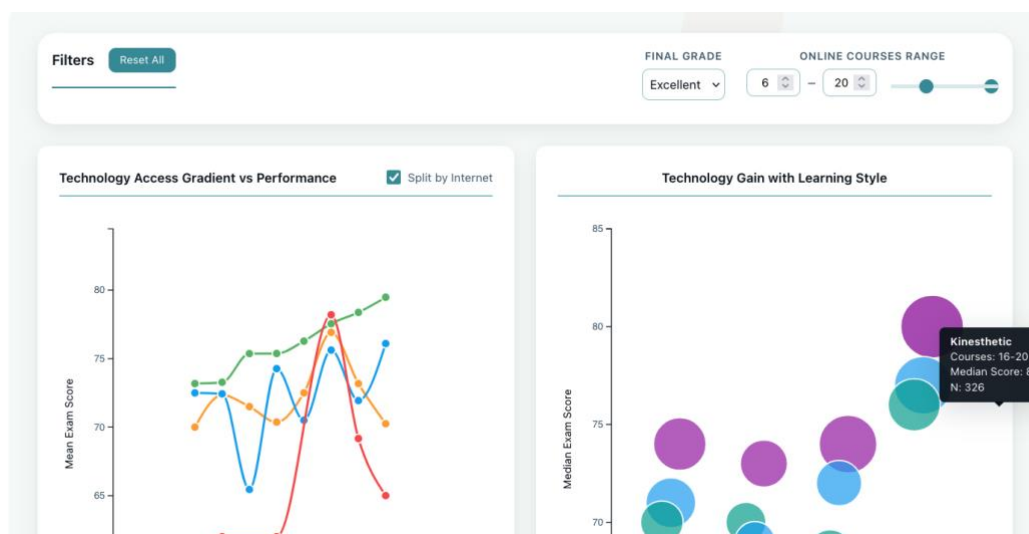
The line chart demonstrates the relationship between the increase in the number of online courses and the changes in exam scores, comparing students with and without education technology access that support discovery of performance trends. Exam scores are mapped on vertical axis (y-axis), while online course participation is mapped on horizontal axis (x-axis), this layout allows users to easily follow the direction of performance over time and determine whether scores increase, decrease, or fluctuate across different levels of online learning engagement. Exam scores for both categories often show an upward trend as the number of students taking online courses increases, but the pattern is more stable and continuously improving for students who have access to EduTech. On the other hand, students who do not have access to education technology show slower progress and have greater fluctuations when they are less engaged online. Users can easily notice how performance is changing over time because the values are always provided on the same scale.

The same line chart also can let users directly make comparisons between students with and without education technology access. Users can compare the results of two groups of students who took the same online course by using different colours to show differences between two groups. By looking at the vertical distance between the two lines, users can identify performance gaps and see how they vary in online engagement increases. When fewer students take online courses, the differences between two groups is smaller, but when more students take online courses, the differences are more obvious. This indicates that students with EduTech access benefit more from higher levels of online learning.



The bubble chart further illustrates how learning styles interact with online learning engagement and technology access by showing variations in median exam scores across different online course participation ranges. Each bubble shows a certain group of learning styles for a certain level of participation in an online course, with horizontal axis (x-axis) indicating the learning style category and vertical axis (y-axis) representing the average exam score. Colour encoding is used to differentiate between high technology and low technology access that allow users to compare performance within each learning style in different technology conditions. The bubble size provides more context about group representation which helps users understand why performance varies based on number of students from each group present.

For most learning styles, the average exam score increases as students take more online courses which shows that there is a positive connection between online participation and academic performance. However, the magnitude of this improvement varies depending on the learning styles. For example, Kinesthetic and Visual learners show higher average exam scores when students participate more in online courses. This indicates these groups may benefit more from having more access to digital learning resources. In contrast, Auditory learners have comparatively low average exam scores at lower and medium involvement levels. As more students participate in online courses, their results gradually improve. Users can examine whether performance patterns are based on large student groups rather than a single student by looking at the different sizes of bubbles, which reflect differences in how groups are represented. By comparing the two visualizations side by side, users may easily spot any irregular patterns and determine if the results are same or different. Users may more easily draw connections between insights and focus on patterns based on the consistent use of colour encoding and clear axis definitions throughout charts. Thus, visual encoding helps with exploration, comparison, and interpretation of data while preserving clarity and avoiding visual complexity.



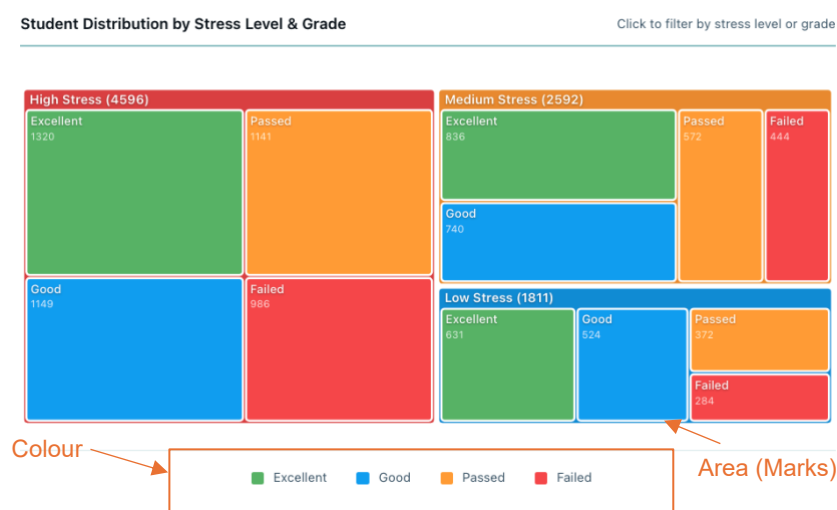
The Task 3 visualisations also implements a dropdown filter for Final Grade and range slider for Online Courses implement select and filter interactions respectively, enabling subset reduction and smooth exploration of continuous attributes. All visualisations feature hover tooltips that implement detailed interaction, providing precise mean and median exam scores on demand. Finally, linked views implement connect interaction where filters affect both charts simultaneously, enabling users to verify technology-performance patterns across different representations.

4.4 Visualizations for Task 4

4.4.1 Marks and Channels

The visualization employs three primary mark types, which differ across visualizations. In the treemap, items are represented as square areas, and the channels for each area are colour hues. For the bubble chart, the markers for items are points, with the x-axis representing motivation level, the y-axis representing academic achievement, colour hue representing stress level distribution, and size representing the number of students. The third stacked bar chart, used line as a mark for the students' proportions, totalling 100 percent. In contrast, the channel uses colour hue to represent the students' academic performance band, x-axis position to represent the participation of students in engagement activities, and length of each part of the line to represent the students' percentage in each academic performance. The summary can be found below:

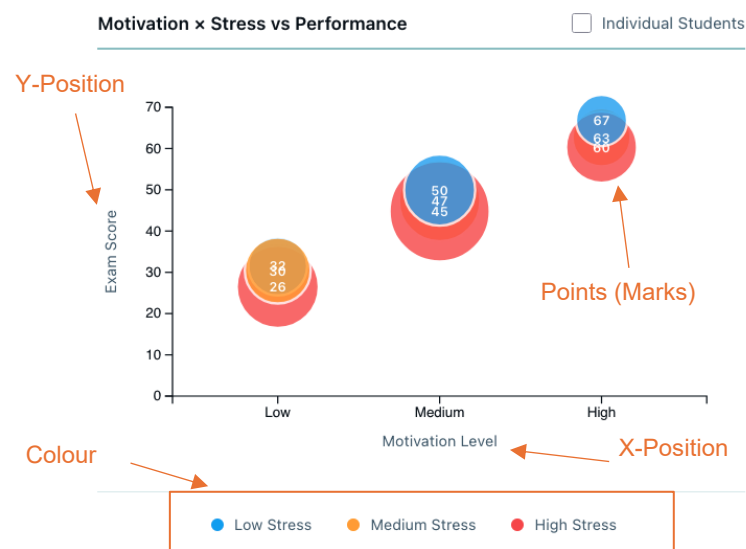
Treemap



Marks: Area

Channels: Colour Hue

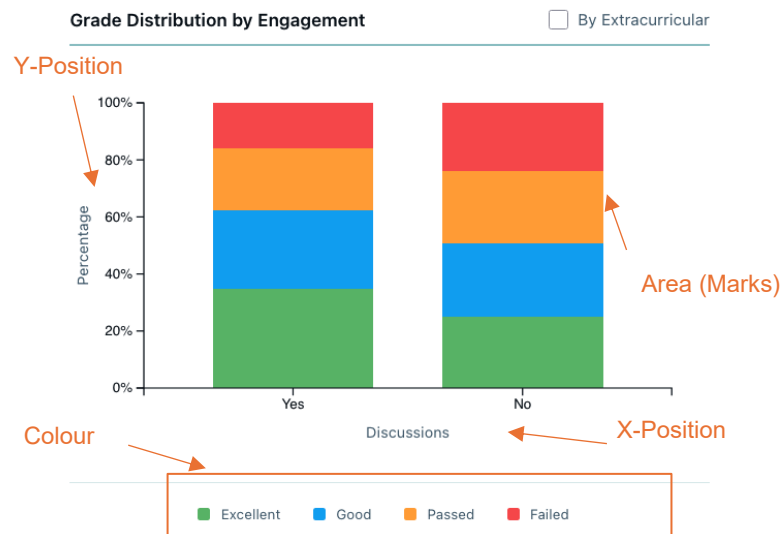
Bubble chart:



Marks: Point

Channels: X-Position, Y-Position, Colour Hue, Size

Stacked Bar Chart

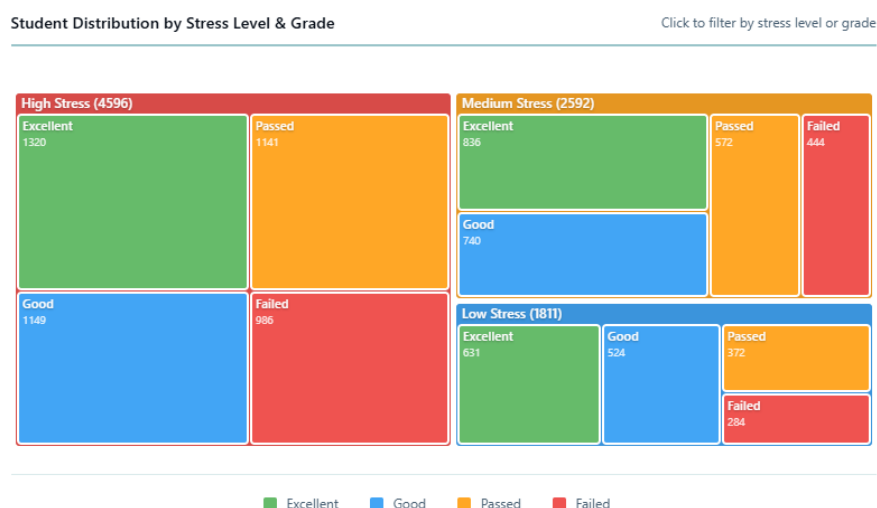


Marks: Area

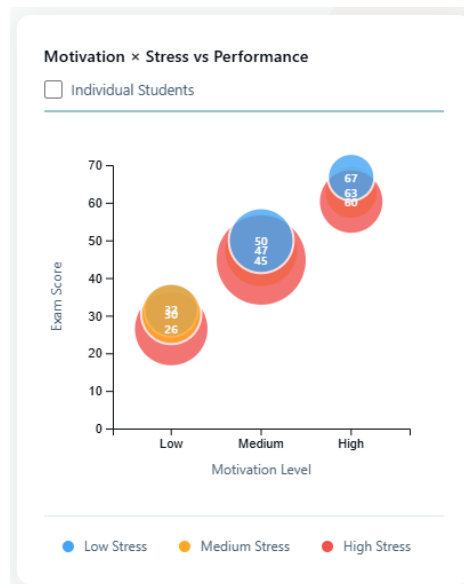
Channels: X-Position, Length, Colour Hue

4.4.2 Visual encodings and analytics

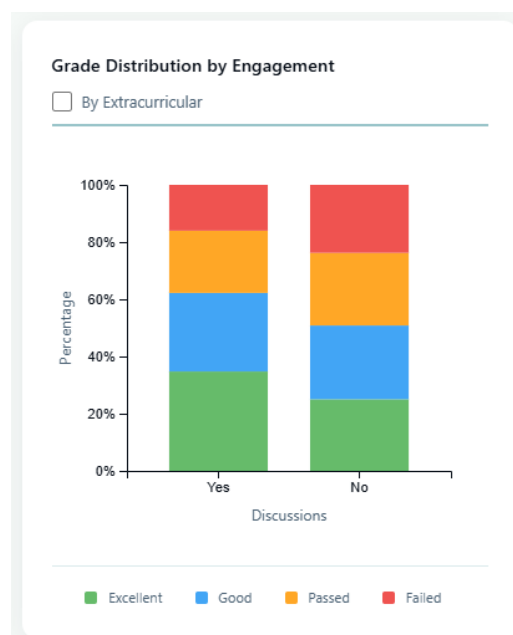
In general, it is not proven that stress level directly affects the academic performance of the students, neither do high-stress students do better, nor do low-stress students do better. As we can see from the distribution, it is balanced across each stress level, with no colour dominating any stress level. The second meaningful information that can be analysed from this visualization is that students in general are experiencing high stress levels, totalling 4596 students, which is higher than medium stress (2592) and low stress (1811) combined. This indicates that students in general face high stress, which does not contribute to better results and is not acceptable, even if it did. Hence, the high stress issue can be identified.



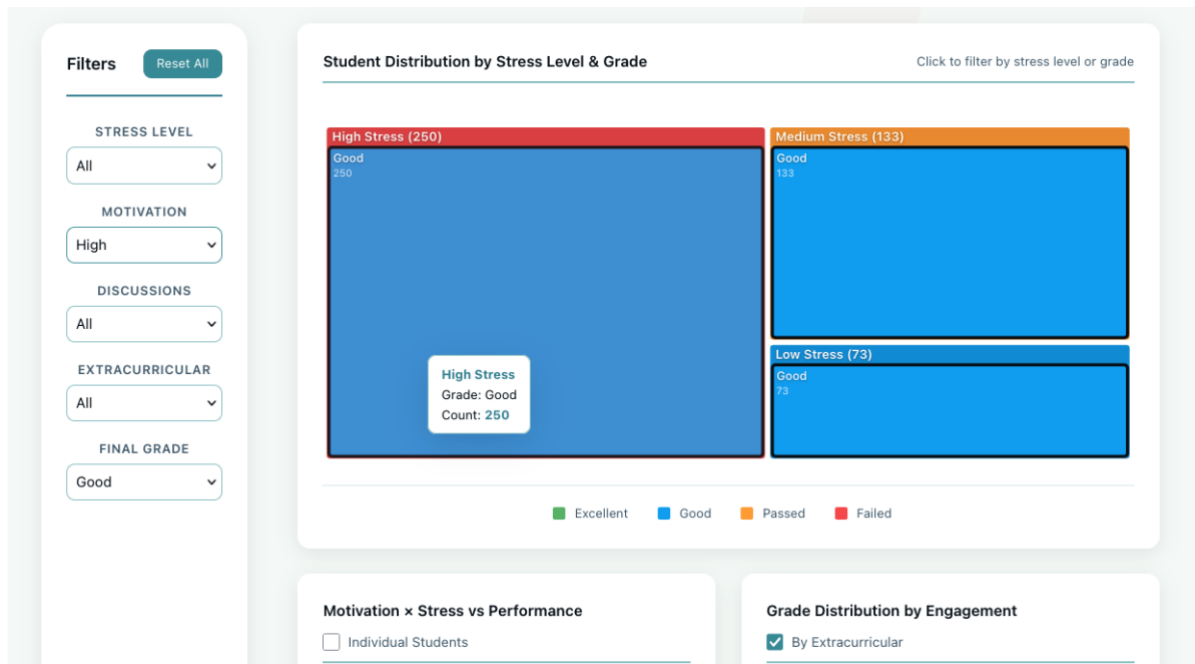
In the second bubble chart, there is a clear distribution difference: highly motivated students achieve higher exam scores, indicating better academic performance. In the meantime, the stress distribution is similar across different motivational levels, with high-stress groups generally at lower positions. However, if we focus on the highly motivated students, it is a subtle cue that low-stress students are actually the top performers within the high-motivation group.



Moreover, under this task abstraction, there are noticeable differences in students' performance across grade levels in participation in discussion activities. Among students who participated in discussions, 34.8% fall into the excellent academic performance band, compared with only 25% among those who did not participate. While participation in discussion helps improve results, there is a similar distribution across groupings by extracurricular participation, indicating it does not affect students' academic performance



In summary, stressing students neither improves their academic performance in the majority, but does provide a subtle cue that low-stress students perform better. The dominant causal factor was student motivation: highly motivated, low-stress students achieved 67 exam scores. Additionally, academic engagement, like group discussion, would help in academic performance, and participating in extracurricular activities does not affect the exam score, for better or worse.



Lastly, the Task 4 visualizations also implements dropdown filters for Stress Level, Motivation, Discussions, Extracurricular, and Final Grade implement select interaction for subset reduction, enabling users to isolate specific psychological and engagement subgroups. The tree map supports direct clicking to filter by stress level or grade, implementing select interaction through visual elements themselves. The "Individual Students" and "By Extracurricular" checkboxes implement reconfigure interaction, allowing users to toggle between aggregated bubbles and individual student scatter points, or switch the stacked bar chart grouping variable. All visualisations feature hover tooltips that implement detailed interaction, displaying precise counts and exam scores on demand. Finally, linked views implement connect interaction where filters affect all three charts simultaneously, enabling users to explore well-being and performance relationships across representations.

5. Four Levels of Validation

Following Munzner's (2014) model for visualization validation, this section documents the validation approaches applied at each level to ensure that the visualisation system effectively addresses the intended domain problems.

5.1 Domain Validation

Domain threats arise from incorrect problem identification at a high level. As no downstream approach is available, it can only be carefully validated by observing and interviewing the target user to accurately pinpoint the needs.

The task definitions and interpretations align with established educational research on the relationships among stress, motivation, engagement, and academic outcomes. The selected tasks reflect real concerns faced by educators and academic institutions. While there are no actual users who can help in validation, like detailed questions and answers, or requirement checks, we fall back on the second choice to ask for peer reviews, where they were asked to take a look at the website our group created, and then provided feedback to see if they were satisfied with the visualization.

Domain validation is the first stage of Muzner's Nested Model and it focuses on visualization and addresses real and unresolved issues in the education domain. Recent study indicates that identifying key predictors of academic success and understanding student performance patterns are ongoing problems in the modern learning environment. For example, a recent study by Manulat (2025) points out the significance of examining student behaviours together with learning outcomes where study habits and school climate are critical and direct predictors of academic success. This highlights that academic success cannot be explained by a single element and must be evaluated within its learning context. Besides, the paper argues that it is ineffective to treat low performing students as a monolith. Meanwhile, educators require tools to differentiate specific behaviours among failing students in order to design targeted interventions (Isman et al., 2024).

Recent study points out the significance of assessing technological access, learning environments, and student well-being, along with academic predictors and performance profiling. The paper shows that inequalities in access to technology can cause performance gaps among students and there is a significant relationship between quality of technology resources and academic performance. They also highlight how limitations in connectivity and digital infrastructure can disadvantage some student groups (Hurtado-Arhuata et al., 2024). The paper highlights the significance of analysing educational technology access as a core domain issue associated with academic equity. In addition, previous study shows that academic stress is a major factor influencing students' mental health and academic performance. Barbayannis et al. (2022) discovered a substantial correlation between academic stress and psychological health among students and highlighted the need of examining stress related patterns for domain level issue identification.

By doing this validation step, we can accurately pinpoint the actual needs. However, if there is additional progress afterward, such as publishing this website, the adoption rate can also be monitored to ensure the problem identification at a high level is correct. This approach ensures that the domain problem is grounded in both evidence-based research and real world relevance, thereby enhancing the credibility of the visualization design.

5.2 Data and Task Validation

Data and task validation focuses on ensuring that the data abstractions and analytical tasks selected are logically compatible, internally consistent, and sufficient to support the intended user reasoning. Tasks were validated by ensuring that each analytical question could be answered directly from the available attributes without requiring external assumptions. This alignment between task intent and data availability is essential to avoid misleading interpretations and to ensure that conclusions drawn from the visualizations remain grounded in the dataset itself.

For data abstraction, answering the ‘what’ regarding the data shown, it should be validated by a proper field study, documenting the actual human usage of the deployed system, so it can be updated to neutralize the threat. In this project, we have a simpler version of this validation, where we cross-check each other’s visualizations within the group to ensure the data shown meets the requirements. Data transformations such as categorization and score normalization were validated by ensuring that we followed established educational analytics standards that allow abstract concepts to be represented as comparable and interpretable values.

Besides that, we verified that each defined task is fully supported by the available data attributes. Quantitative attributes such as exam scores, online course participation, and assignment completion directly help in finding correlations and trends. While categorical attributes including demographic groupings, technology access, and learning style enable meaningful group comparisons. This shows that no required inferred task or missing attributes which reduce the risk of unsupported analysis. Synthetic data constraints were explicitly acknowledged by one of the group members during visualization creation, and a discussion was held to decide whether we should proceed with what we had. At that point, we found a new set of datasets as a backup, in case the worst happened. At the end, we consulted the supervisor and proceeded to filter the dataset, which finally allowed us to extract meaningful information.

For task abstraction, answering the ‘why’ of the data shown, it could be validated by an immediate approach which is testing the deliverables or the protocol with the target users. Then, we collect evidence of utility from the visualization. This action intends to confirm if the visualization fits the needs and requirements. Similarly, due to the time constraints in this project, the best we can do for the task validation is to recheck and re-evaluate each member's task abstraction and product visualization, making sure they fit the need. Each task was designed to support different types of user reasoning such as identifying key influencing factors, exploring conditional relationships, comparing groups, etc. Although different tasks may use the same attributes, they provide different analytical goals, thus each task contributes unique insight rather than duplicating findings.

Other than that, as the project did not reach a deployment stage, no downstream or post-deployment validation was conducted. Nevertheless, the applied validation steps provide reasonable assurance that the data abstractions and task designs are coherent, justified, and appropriate for the intended analytical objectives.

5.3 Design Validation

Idiom threats occur when the implementation fails. Logically, both the immediate and downstream approaches are possible to fix the danger. Following Munzner’s effectiveness ranking, our design validation prioritized perceptually accurate encodings for the most important comparisons, and then verified whether users could interpret and interact with the views as intended through lightweight user testing. In the case of a base technical error coming from faulty code or implementation, an immediate approach should be taken to validate the

encoding and interaction design. In cases of higher-level mistakes, such as data collection, a downstream approach should be used to verify that the data is organized.

In general, some of the product visualization was not satisfying, as we were working on Power BI from the start, which was not user-friendly, and some colour encoding choices were not explicitly selected. Moreover, we are working separately and there is not enough consistency across the charts, either. So our group leader took on the responsibility of consolidating all the view products into a consistent chart that adhered to established visualization principles, using colour for categorical distinction, position for comparison, and so on. This way, we can improve interpretability and task efficiency.

The most crucial quantitative attributes such as exam scores and online course participation were consistently used to encode using position on a common scale throughout the final integrated visualizations that allow for precise comparison. To prevent interference with quantitative judgements, colour hue was reserved strictly for categorical distinctions, including technology access and group membership. Interactive elements such as filtering and coordinated views were also applied to reduce cognitive load by allowing users to focus on relevant subsets of the data without overwhelming the visualization. The final design follows established visualization principles that help users interpret the data and complete the intended tasks more effectively.

5.4 Algorithm Validation

Algorithm threats arising from slow algorithm implementation should be validated by analysing computational complexity, and the system time/memory can then be further measured once the pinpointed optimization opportunities are validated and fixed. So in this stage of validation, we address whether the computational implementation is correct and performant enough to support the chosen techniques.

This project does not include any complex predictive algorithms, which reduces the risk of analytical bias that are commonly associated with machine learning approaches. Most computation in this project involves aggregation, grouping, and summary statistics such as means, counts, percentages, and correlation coefficients. Therefore, algorithm validation focused only on two aspects, correctness of derived calculations and responsiveness of interaction updates with the visualizations.

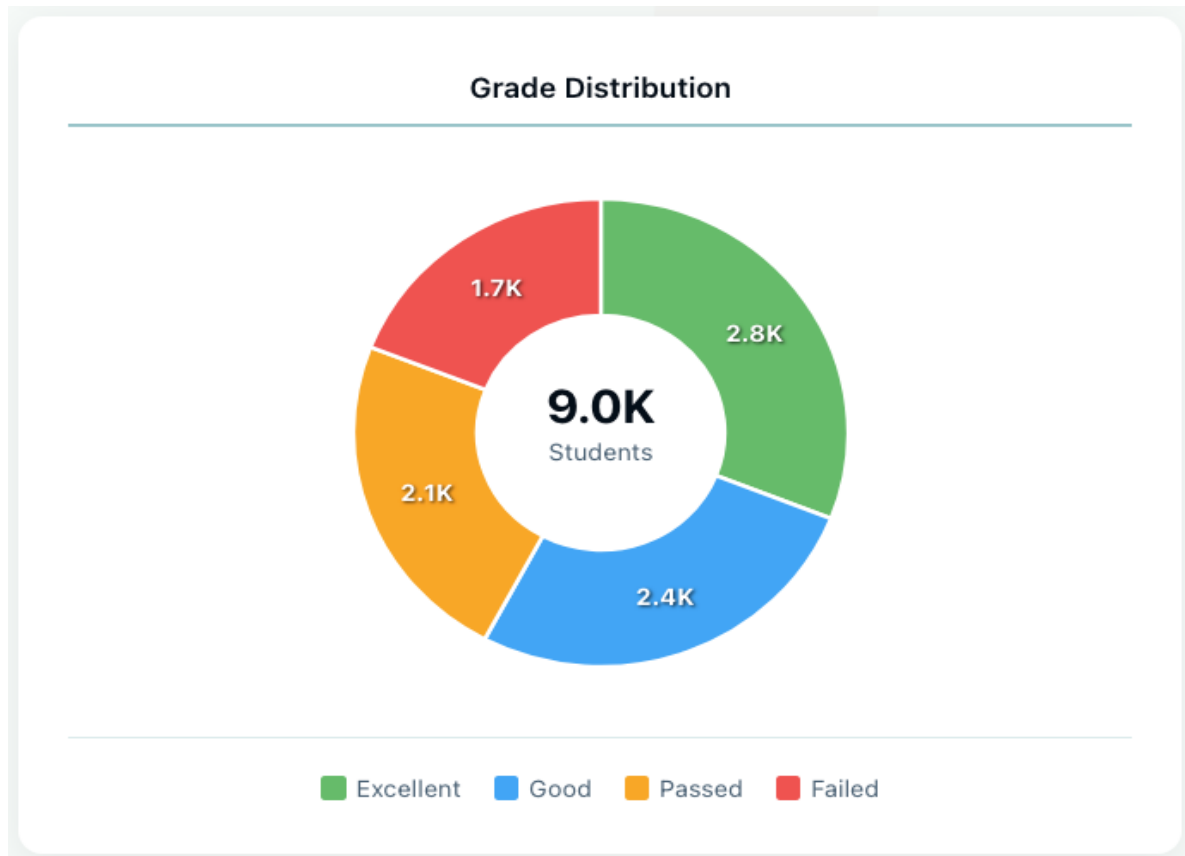
The validation of the algorithm was through cross-checking across different visualizations. The aggregation logic was kept consistent across different charts and derived values were also compared across different views to ensure that they matched when filtered to equivalent subsets. For instance, the grade distribution counts shown in the donut chart were verified to make sure the sum is correct to the total student count and match the stacked histogram totals. Similarly, average values displayed in the line charts were cross-referenced with the underlying grouped calculations to confirm accuracy. Correlation calculations were also checked for consistency with the displayed values and the proposed data encoding.

Besides that, performance validation was based on the practical testing during development. With about 9,000 rows of records, the implementation of D3.js JavaScript library were able to load the static webpage under 3 seconds on standard home broadband connection and update the linked views responsively under typical usage such as filtering, hover queries and basic interactions without noticeable latency.

6. Final Visualisations

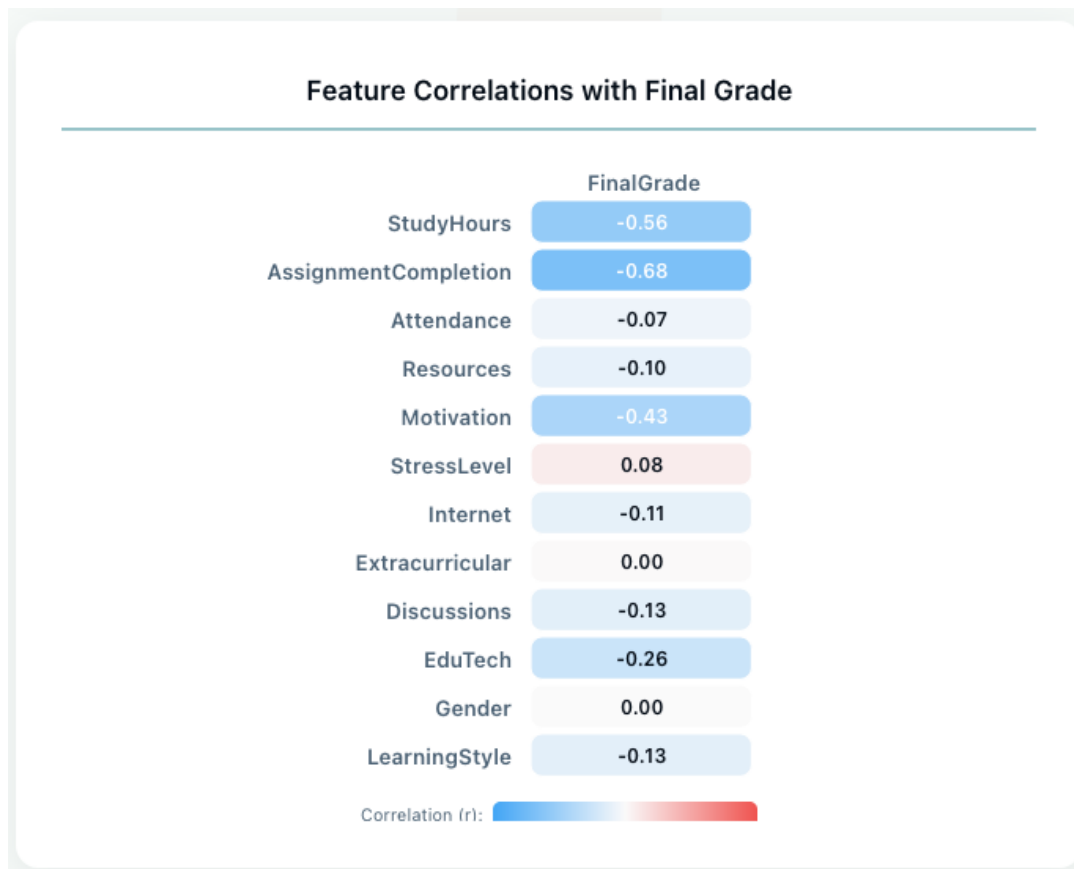
Figure 1

Grade Distribution



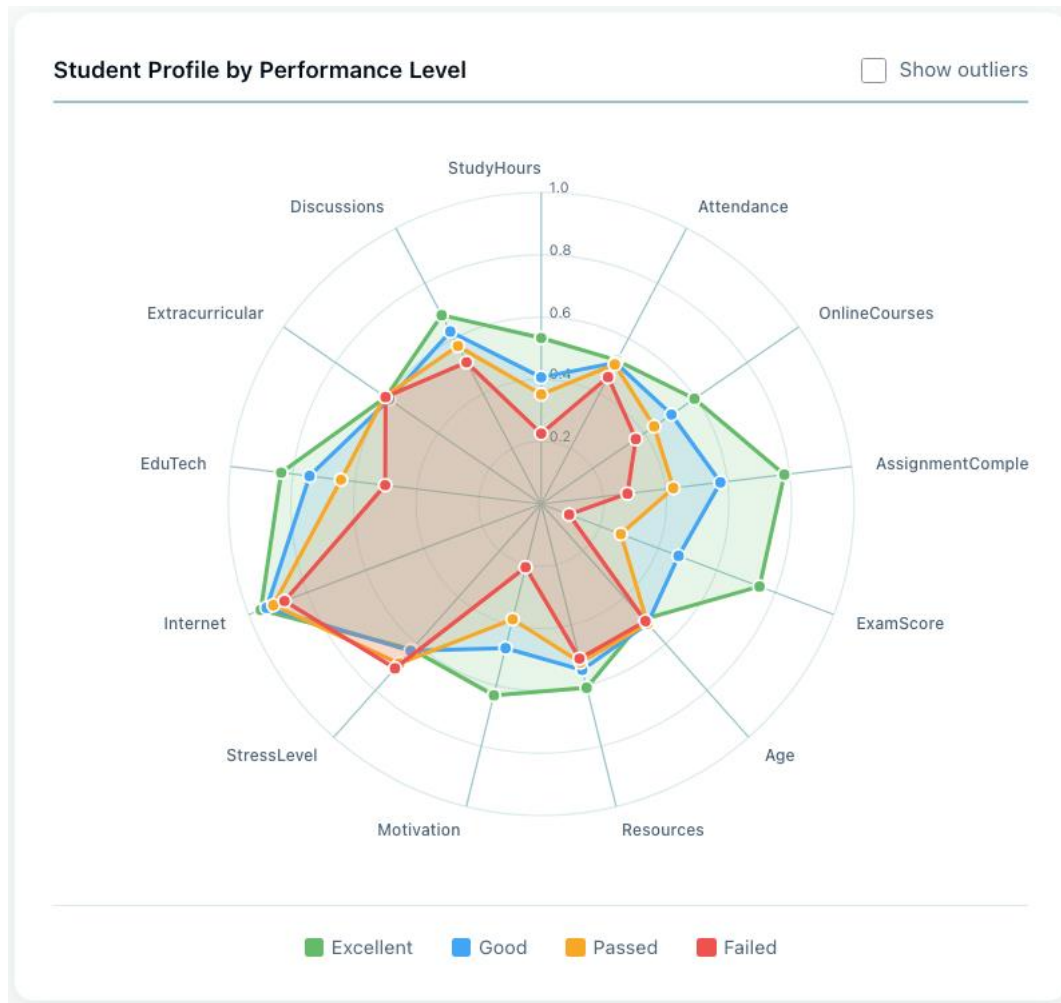
Note. The figure illustrates the distribution of academic performance among 9,000 students. Grades are categorized as Excellent ($n = 2,800$), Good ($n = 2,400$), Passed ($n = 2,100$), and Failed ($n = 1,700$). Values represent student counts, and percentages may not total 100% due to rounding.

Figure 2

Feature Correlations with Final Grade

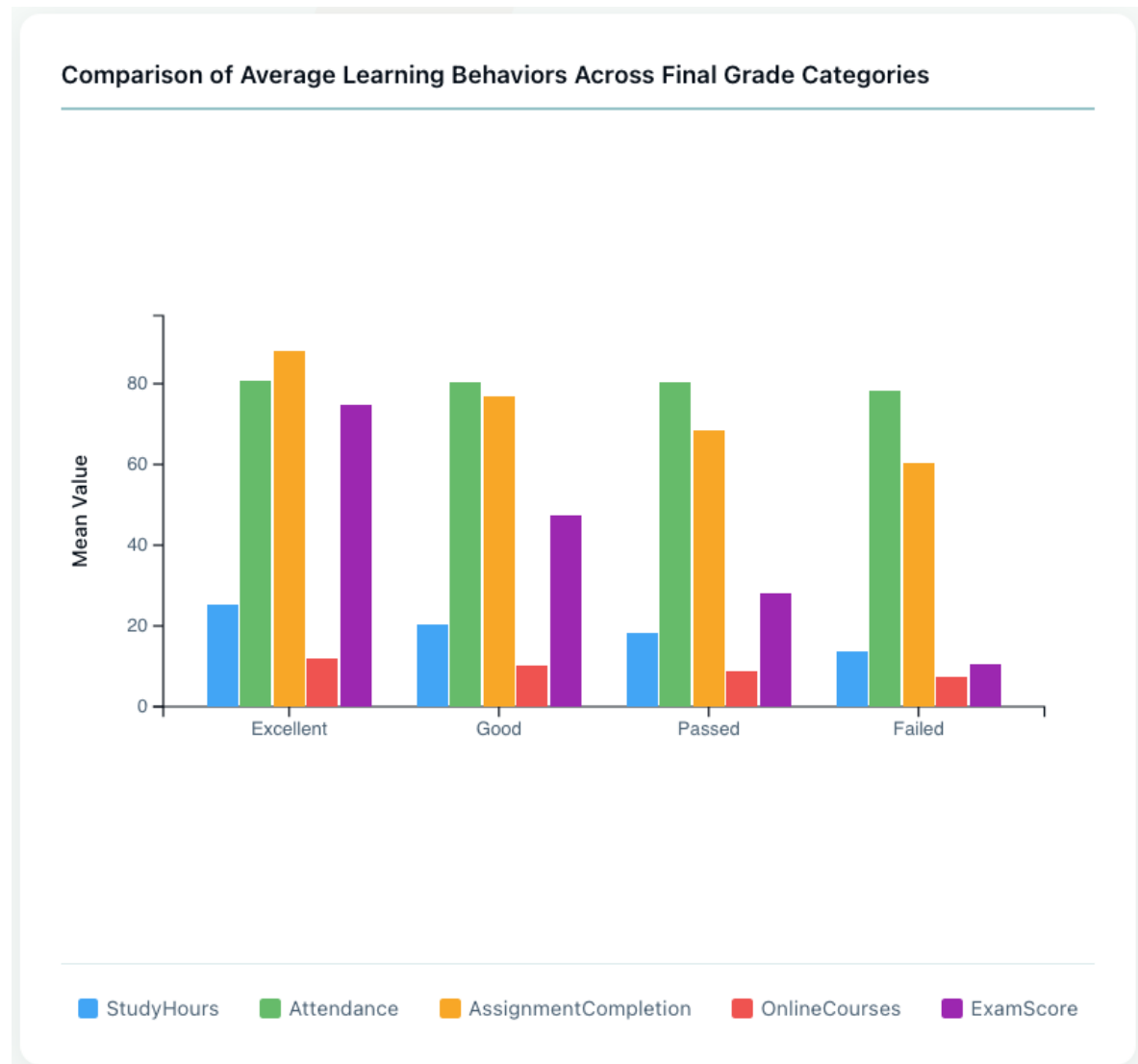
Note. The figure presents Pearson correlation coefficients (r) between selected student-related features and FinalGrade. Negative correlation values indicate that higher levels of the feature are associated with better academic outcomes, as lower FinalGrade values represent higher achievement categories. Positive values indicate an association with poorer outcomes. Correlation strength varies across features, with assignment completion, study hours, and motivation showing the strongest relationships.

Figure 3

Student Profile by Performance Level

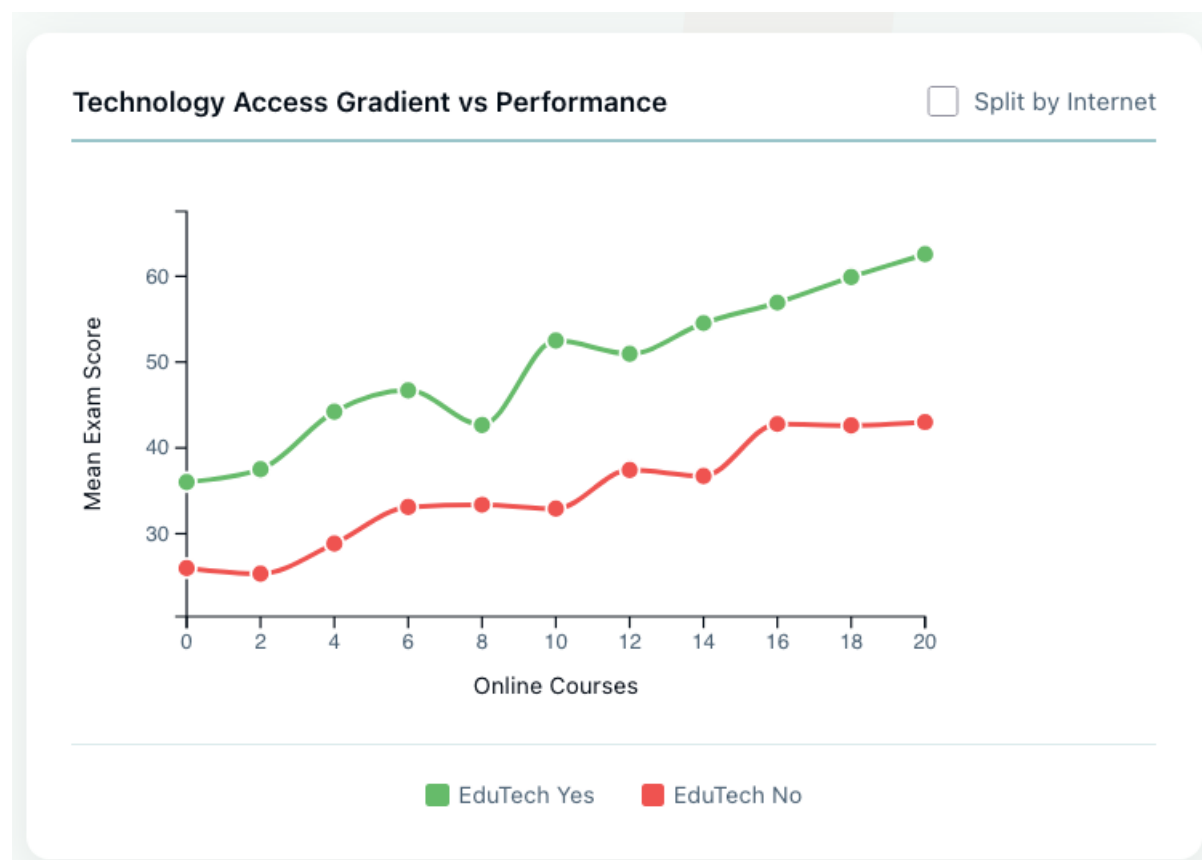
Note. The figure compares normalized mean values of academic, behavioural, and environmental attributes across student performance levels (Excellent, Good, Passed, Failed). All variables are scaled to a common range from 0 to 1 to enable relative comparison across dimensions. Larger radial values indicate higher average levels of the corresponding attribute. The visualization highlights systematic differences in study habits, engagement, and resource access across performance groups rather than individual student variation.

Figure 4

Comparison of Average Learning Behaviours Across Final Grade Categories

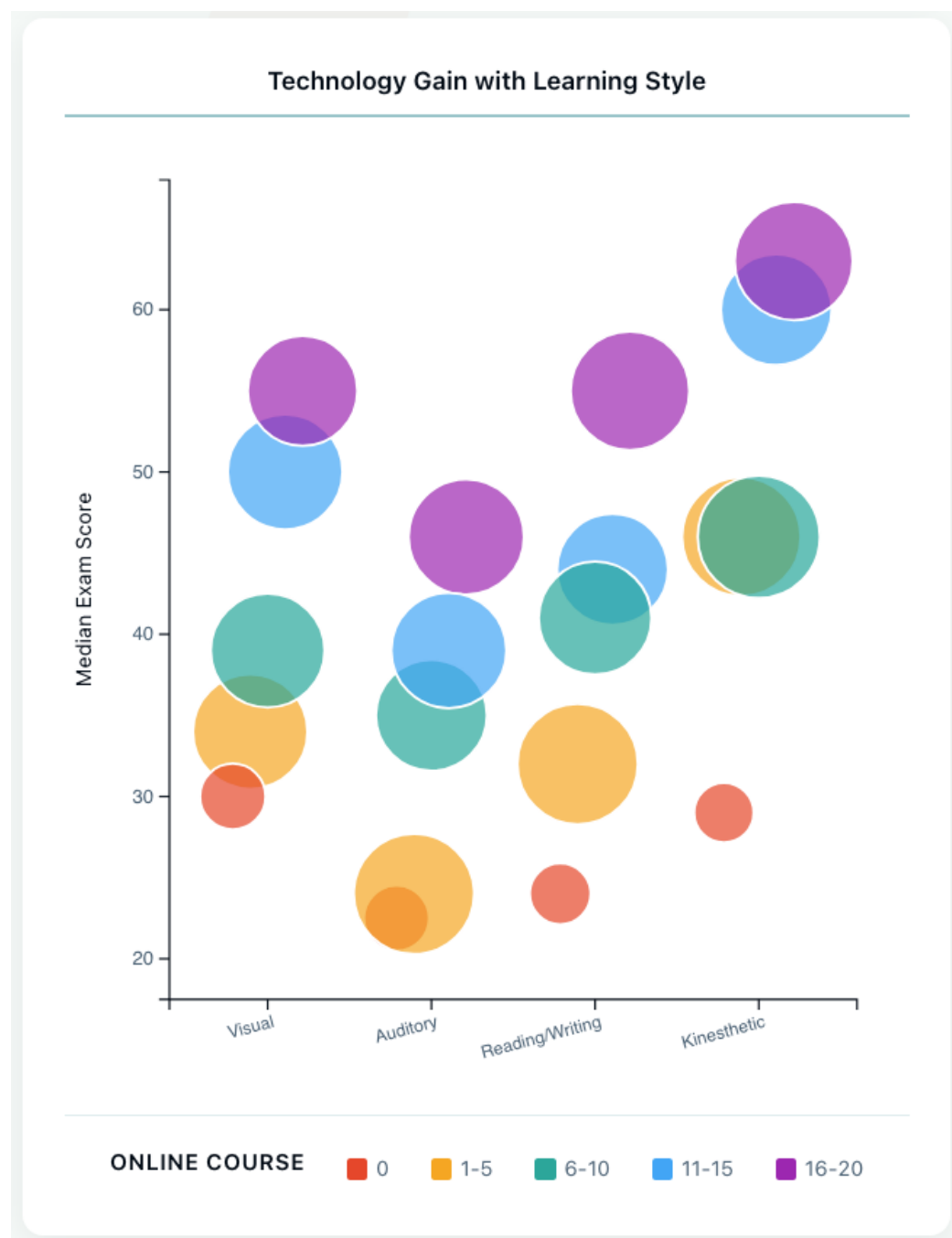
Note. The figure compares mean values of selected learning behaviour variables across final grade categories (Excellent, Good, Passed, Failed). Higher bar values indicate greater average levels of the corresponding behaviour. The comparison is descriptive and intended to illustrate systematic differences in study time, attendance, assignment completion, online course participation, and exam performance across achievement groups; no causal inference is implied.

Figure 5

Technology Access Gradient vs Performance

Note. The figure illustrates the relationship between the number of online courses accessed and mean exam scores, stratified by educational technology usage, EduTech Yes vs. No. Mean exam scores generally increase with greater online course engagement, with consistently higher scores observed among students with EduTech access. The trends are descriptive and indicate association rather than causation.

Figure 6

Technology Gain with Learning Style

Note. The figure depicts median exam scores across learning styles under varying levels of online course participation. Bubble size represents relative student counts within each category, while colour encodes the range of online courses completed. Higher median exam scores are generally associated with greater online course engagement, with kinaesthetic learners showing the highest median performance across most participation levels. The visualization is descriptive and does not imply causal relationships.

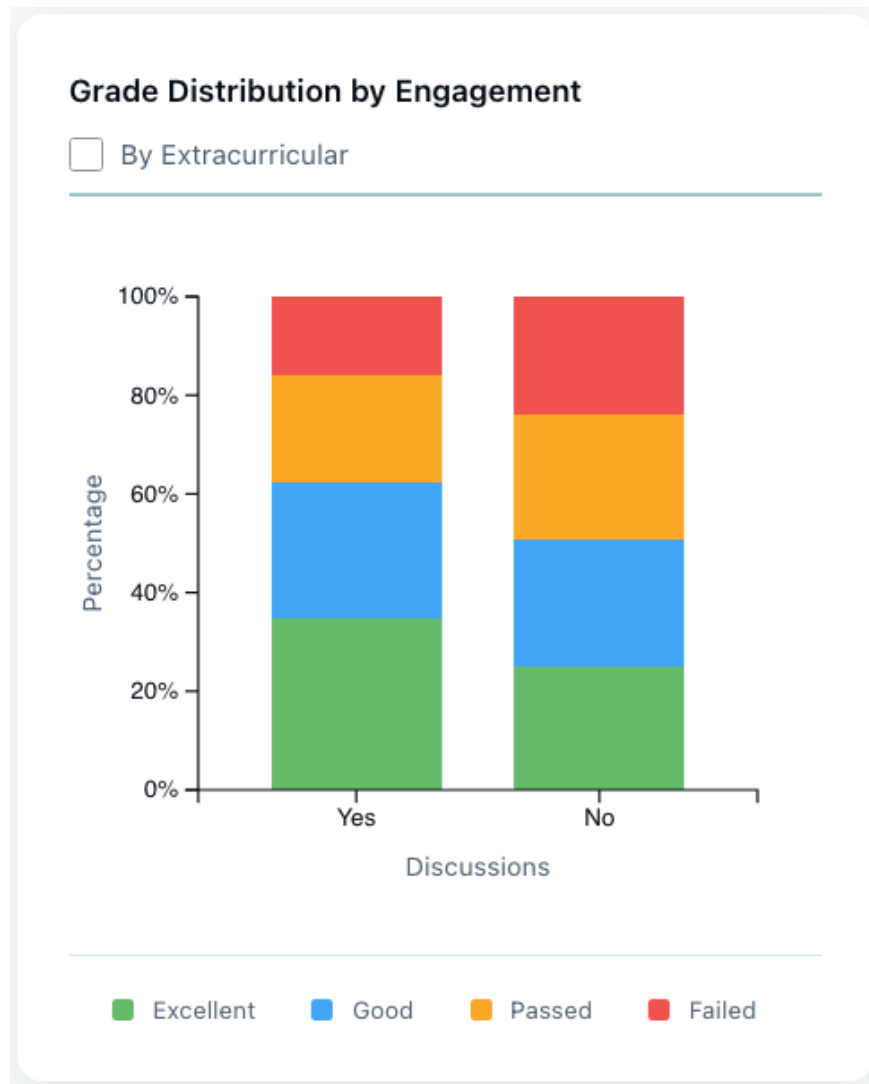
Figure 7

Student Distribution by Stress Level and Grade



Note. The figure presents the distribution of students across stress levels and final grade categories using a treemap. Rectangle size corresponds to the number of students within each stress-grade combination, while colour denotes grade category. The visualization highlights that high stress is the most prevalent condition across the dataset, yet substantial proportions of high-stress students still achieve good or excellent outcomes. Results are descriptive and intended to support comparative analysis rather than causal inference.

Figure 8

Grade Distribution By Engagement

Note. The figure shows the percentage distribution of final grade categories by class discussion participation. Stacked bars represent the proportion of students achieving Excellent, Good, Passed, and Failed outcomes within each engagement group. Students who participate in discussions exhibit a higher combined proportion of Excellent and Good grades and a lower proportion of failures compared to those who do not participate. The visualization is descriptive and does not imply causality.

7. Discussion and Reflection

7.1 What Worked

For this project, some design and implementation choices worked well in supporting an easy to understand and engaging data story telling. First, the website followed a narrative flow that went from context framing then to the evidence and interpretation. The overall structure of the website aids understanding on why the section is shown to them and how each of these visualizations can show them the argument rather than showing the charts as isolated outputs with no story to tell. The framing of the narrative also supported good pacing with the early section bringing up expectation and terminology, while the middle section asked users to explore and the later sections consolidate the findings into takeaways that will make users feel connected to what they had seen from the previous section.

Besides that, interactivity in the website helped improve the experience by letting users control how they want to explore the data. Filters and hover tooltips let users check details on demand and let them focus on specific subset without needing to look for another static chart for every scenario. This type of interaction allows exploration where users can test if the pattern remains consistent when they choose a subset with a certain group. In practice, this lowers the risk that users might overgeneralize data from a single aggregated view just because the interface made it easy to validate the patterns and distribution across different categories.

In addition, using coordinated views with linked charts made it easier to see relationships from more than one angle. Rather than relying on a single chart to explain everything, the mini dashboards brought together multiple views. For example distributions, group comparisons, and the relationships between variables. In this case users could build a fuller and clearer understanding easily. This works especially well for educational data, where performance is usually influenced by several factors at once. Seeing the same categories reflected across multiple charts will also encourage users to cross check what they are observing, which is important in helping users to prevent conclusions based on one potentially misleading view.

Finally, we found that accessibility choices had improved overall usability. Grade categories were encoded consistently across the visuals, and then the color palette was selected to remain readable for users with color vision deficiencies. Instead of being just a “nice-to-have,” these decisions support understanding by making category distinctions clearer and reducing reliance on subtle color differences as well. The website also uses animation in order to make the narrative more engaging. In the meanwhile, the smooth transitions and the small illustration movements have created a better visual flow and help maintain attention as users move through the story.

7.2 What Didn't Work

The largest challenge was the dataset itself, the data is synthetic or transformed. It limited how confidently the story could claim to represent real student behaviour. Even when visual patterns appear coherent, synthetic data can flatten distributions, reduce natural noise, or unintentionally create relationships that look “too clean.” This created a practical tension: the visualizations could be technically correct and visually readable, but the interpretation still needed to remain cautious to avoid overstating implications.

A second challenge involved some limitations in certain chart choices. As we know, radar charts were useful for quick comparison, but they are weaker for precise value reading because users must judge angle and radial distance rather than position on a common scale. This trade off was acceptable when the radar chart served as an overview paired with more precise charts, but it still introduced a risk that users might rely on viewing the shape rather than just the numeric detail. Similarly, the dense views can increase users' cognitive load, especially for average audiences, even when they provide comprehensive information.

A third challenge was responsiveness and screen constraints. Multi-view dashboards work best when the user has enough space to compare charts side by side. On mobile layouts, the design inevitably shifts toward vertical stacking, which reduces immediate cross-chart comparison and can increase scrolling fatigue. Maintaining clarity required trade-offs such as simplifying labels, adjusting chart size, or reducing the number of simultaneously visible views.

7.3 Limitations

Several limitations should be stated explicitly. The project uses a synthetic dataset, which may not reflect real student distributions. As a result, the story should be treated as a demonstration of visual analytics design and reasoning rather than a definitive statement about student performance. The analysis is also correlational, meaning that even strong relationships in the charts do not establish causation. For example, a strong association between assignment completion and grades does not prove that increasing completion alone will cause grade improvements; both may be influenced by underlying traits such as discipline or prior preparation. If the data were collected in real settings, some variables such as stress, motivation and study hours would likely be self-reported and subject to response bias and measurement noise.

There were also some limitations about the visualization. In some of the charts, there was minor clutter where marks stacked too closely, most notably in the Task 4 bubble chart. This was mitigated by adding text encoding to indicate the number of students in each group, which improved clarity and interpretability. However, the need for mitigation on this visualization indicates a general constraint, when many points compete for limited space, the overlapping can reduce readability unless aggregation or text encodings are applied.

Another limitation is that many of the insights come from aggregated summaries such as group means and normalized averages, which can hide how much students vary within the same performance group. If groups overlap a lot, small gaps in the averages may not be meaningful, and without showing distributions, uncertainty, or group sizes, users might treat the patterns as more consistent than they really are. This matters most for the profile comparison views. In the radar chart, adding many attributes makes the lines harder to compare because they overlap, and turning on outliers adds even more visual clutter. Normalization helps users compare attributes on the same scale, but it also removes in the real world meaning of the values. The grouped bar chart is clearer for comparing groups within a single attribute, but as each attribute has different units and ranges, it still can lead to incorrect comparisons if someone reads bar heights across different measures as if they were directly comparable.

7.4 Future Improvements

Future iterations can improve both validity and usability. From a data perspective, the most valuable enhancement would be incorporating real or institutionally collected data with appropriate privacy safeguards, or at minimum validating patterns with a more realistic distribution. Adding attributes that track students across time would also strengthen the narrative because it enables questions about change and progression rather than relying only on single snapshots of data. This would support more meaningful interpretations, such as whether early behaviours predict later outcomes or whether improvements in engagement correspond to improved performance.

From a visualization perspective, clutter reduction should be addressed more systematically. For bubble and scatter-style charts, aggregation like binning and jittering could reduce overlap without affecting the pattern or distributions. Mobile optimization can also be improved by using progressive disclosure, presenting a simple headline chart first, then allowing users to expand details on demand. This preserves the strength of multi-view analysis while adapting to smaller screens. Finally, adding some short interviews from experts could strengthen the story by explaining why patterns appear, bridging the gap between statistical relationships and real learning experiences for average users.

A further improvement is to strengthen multivariate profile interpretation in a way that remains readable at scale. When profile style charts include many attributes, allowing users to select, group, or collapse attributes can reduce overlap and make comparisons clearer. Adding an optional distribution view for example a small multiples of boxplots or density strips by group would also help users assess whether differences in group averages reflect broad shifts or are driven by a subset of students.

From an analytics perspective, adding in a machine learning model could transform the system from data explanation to a much more useful decision support system. A lightweight model could identify high risk student profiles early and present results with interpretability features such as feature importance, confidence indicators so users understand what drives the predictions.

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1. Introduction and Background

2. Data Abstraction

3. Task Abstraction

Task 1: Identify Key Predictor of Academic Success

Task 2: Compare Student Profiles Across Performance Levels

Task 3: Explore the Impact of Learning Environment and Technology

Task 4: Analyse Well-being and Engagement Factors

4. Design Justification

5. Four Levels of Validation

6. Final Visualization

7. References

Appendix